Jackknife Variance Estimation for Hájek-Dominated Generalized U-Statistics

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Abstract: We prove ratio-consistency of the jackknife variance estimator, and certain variants, for a broad class of generalized U-statistics whose variance is asymptotically dominated by their Hájek projection, with the classical fixed-order case recovered as a special instance. This Hájek projection dominance condition unifies and generalizes several criteria in the existing literature, placing the simple nonparametric jackknife on the same footing as the infinitesimal jackknife in the generalized setting. As an illustration, we apply our result to the two-scale distributional nearest-neighbor regression estimator, obtaining consistent variance estimates under substantially weaker conditions than previously required.

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1. Introduction

Generalized U-statistics, introduced by [11], unify a variety of modern extensions to the classical theory of U-statistics: randomized, incomplete, and infinite-order U-statistics, within a single analytical framework. These estimators include a wide range of random forest (RF) learners, many of which are of growing interest in nonparametric inference. While some theory is now available concerning their asymptotic behavior, variance estimation for this class of statistics remains less well understood. In many applications, the jackknife remains popular for variance estimation due to its simplicity, but theoretical guarantees outside the classical fixed-order setting are comparatively scarce.

In this paper, we establish conditions under which the jackknife yields ratio-consistent variance estimates for generalized U-statistics. Our main insight is that the consistency of the jackknife hinges not on the full structure of the statistic, but rather on a simple condition: namely, that the variance of the statistic is dominated by that of its first-order (Hájek) projection. This criterion, originally developed to justify central limit theorems for generalized U-statistics, naturally carries over to variance estimation and enables a unified treatment of the jackknife and its delete-d variants. In particular, we extend existing results from the classical literature on U-statistics (e.g., [2]) to the generalized setting without modifying the base estimator. Our analysis proceeds by tracking the contribution of higher-order Hoeffding projections and showing that their cumulative influence vanishes under asymptotic Hájek dominance. These results help close a conceptual and practical gap between the jackknife and the infinitesimal jackknife (IJ), the latter of which has seen wider adoption in the context of RF inference. Moreover, the criteria we impose are milder than many found in prior work, enabling broader applicability in practice.

To illustrate this point, we revisit the Two-Scale Distributional Nearest-Neighbor (TDNN) regression estimator of [6]. We show that under mild regularity conditions, this estimator

satisfies asymptotic Hájek dominance in regimes where the kernel order grows at a slower rate than the sample size, substantially loosening the original assumptions needed to justify the use of jackknife-based variance estimates.

Related Literature

Historically, these results are based on the well-established theory concerning U-statistics starting with [7]. The classical results built on this seminal work have been a well-established tool in mathematical statistics for a long time. Thus, there is a significant body of literature that studies their properties, including outstanding introductions such as [8]. Concerning variance estimation for U-statistics, two highly related papers are the aforementioned [2], exploring the theory of the jackknife when applied to U-statistics, and [1] which fulfills a similar role for the bootstrap. While relatively new, there is a significant body of literature concerning infinite-order U-statistics, which share their structure with the TDNN estimator, including results concerning variance estimation. In particular, [16] proposed jackknife and IJ estimators for RF prediction variance and demonstrated their practical appeal in large ensembles, thereby helping to popularize IJ-style variance estimation in this context. [10] take a different route: exploiting the U-statistic representation of subbagged/random-subsample ensembles, they form structured (nested) Monte Carlo estimators for the leading Hoeffding term and derive pointwise CLTs; crucially, their internal estimator delivers the needed variance components with essentially no extra computational cost beyond fitting the forest under small-subsample regimes. Our use of Hájek dominance clarifies that this line targets the same first-order component our theory designates as dominant, albeit with an explicit MC estimator rather than jackknife-type resampling. Building on these ideas, [18] recast the problem in a with-replacement (V-statistic) framework and propose the balanced method. They diagnose why many corrections (including IJ-style ones) can be upward biased or even yield negative variance estimates unless the ensemble is taken much larger than needed for prediction, and argue that subsampling with replacement mitigates this sensitivity; the balanced method then offers a computationally simple estimator with reduced bias for a fixed ensemble size. Most recently, [17] revisits variance estimation for generalized U-statistics, including RFs, through what they coin a peak-region dominance perspective and introduce a matched-sample estimator designed to mitigate bias when the subsample size s is not small relative to n. They document the bias tendencies of leading-term-dominance methods when s is large, and show that their variants remain stable and largely unbiased across a range of s. The most directly relevant development for the present work is [12], which reframes IJ through several equivalent lenses (including a regression view), and gives a careful biasconsistency analysis. Most pertinent to subsampling-based generalized U-statistics, they provide a consistent IJ-for-U-statistics construction called the pseudo-infinitesimal jackknife under conditions compatible with Hájek-style projection dominance, thereby bridging IJ ideas to the subsampling world that underlies modern RF-style estimators.

Our contribution sits precisely in this projection-dominance lane: we show that once the variance of a generalized U-statistic is asymptotically governed by its Hájek projection, the plain jackknife (and its delete-d variants) is ratio-consistent without modifying the estimator or imposing strong smoothness assumptions; we also extend the argument to incomplete

(Bernoullisampled) settings under what we call an asymptotically sufficient sampling condition. As the purpose of variance estimation in the problem at hand is ultimately to employ distributional approximations, papers such as [5] and [13] are similarly of high relevance for potential applications. Due to the close connection to the RF method introduced by [4], there is also a relevant overlap with the literature on that topic. Thus, articles such as [15] are of special interest, especially since causal forests are considered the state-of-the-art technique for estimating heterogeneous treatment effects.

Notation

Throughout this paper, we will use a number of notational conventions: some standard in the U-statistics literature, some introduced to allow for convenient notation in the specific domain that is addressed. First, let $[n] = \{1, \ldots, n\}$. Given a finite index set $\mathcal{I} \subset \mathbb{N}$, we introduce the following notational conventions.

$$L_s(\mathcal{I}) = \{(l_1, \ldots, l_s) \in \mathcal{I}^s | l_1 < l_2 < \ldots < l_s\} \quad \text{and} \quad L_{n,s} = L_s([n]) \tag{1.1}$$

We will denote a data set consisting of n observations $(Z_1, \ldots, Z_n)$ drawn i.i.d. from some distribution F_Z by $D_{[n]}$. We will denote the measure associated with F_Z by μ_Z, that is $D_{[n]} \sim \bigotimes_{i=1}^{n} \mu_Z$. A realization of such a data set will be denoted by $d_{[n]}$. For a data set $D_{[n]}$ and a vector $\ell \in L_{n,s}$, denote by $D_{[n],-\ell}$ the data set where the observations corresponding to indices in ℓ have been removed. To simplify the notation in the case that a single observation (say the $i'th$ observation) is removed, we use the notation $D_{[n],-i}$. Similarly, given such a data set $D_{[n]}$ and index vector ℓ, denote by D_ℓ the data set consisting only of the observations in $D_{[n]}$ corresponding to the indices in ℓ. In analogy, we will use similar notation for the covariates and responses, e.g. $X_{[c]}$ to refer to $(X_1, \ldots, X_c)$ and $Y_{[c]}$ for $(Y_1, \ldots, Y_c)$. We will analogously define the corresponding notation for a realization of the data set $d_{[n]}$ and its components $x_{[n]}$ and $y_{[n]}$. In an abuse of notation, when considering two index vectors ℓ and ι that do not share any entries, we denote by $\ell \cup \iota$ the concatenation of the two vectors, e.g., if $\ell = (8, 2, 5)$ and $\iota = (1, 6)$, then $\ell \cup \iota = (8, 2, 5, 1, 6)$. In the following, $\rightsquigarrow$ denotes weak convergence, while $\longrightarrow_p$ denotes convergence in probability. We will use the symbol $\lesssim$ to denote an inequality that holds for sufficiently large sample sizes n and associated kernel orders s. As we consider settings where these diverge together, the specific reference parameter will be clear from the context.

2. Generalized U-Statistics

Throughout this paper, we will use the framework of generalized U-statistics as introduced in [11]. This framework unifies the existing concepts of incomplete, randomized, and infinite-order U-statistics. It thus covers popular estimators such as random forests and, more generally, a broad class of ensemble estimators. A mature statistical theory of generalized U-statistics could, therefore, provide the foundation for the widespread usage of these estimators in fields such as computer science or economics.

Definition 2.1 (Generalized U-Statistic)**.**
Suppose $D_{[n]} = (Z_1, \ldots, Z_n)$ is a data set consisting of i.i.d. observations from F_Z. Let h

denote a (possibly randomized) real-valued function utilizing s of these observations that is permutation-symmetric in those s arguments. A generalized U-statistic with kernel h of order (rank) s refers to any estimator of the form

$$U_{n,s,N,\omega}\left(D_{[n]}\right) = \frac{1}{\widehat{N}} \sum_{\ell \in L_{n,s}} \rho_\ell h\left(D_\ell; \omega\right) \tag{2.1}$$

where ω denotes i.i.d. randomness, independent of the original data. $(\rho_\ell)_{\ell \in L_{n,s}}$ denotes a collection of i.i.d. Bernoulli random variables determining which subsamples are selected and is independent of all other inputs to the U-statistic. Furthermore, we define $p \coloneqq \Pr(\rho_\ell = 1) = N/\binom{n}{s}$ and say that the actual number of selected subsamples is given by $\widehat{N} = \sum_{\ell \in L_{n,s}} \rho_\ell$ where $\mathbb{E}\left[\widehat{N}\right] = N$.

- When $N = \binom{n}{s}$, the estimator in Eq. 2.1 is a complete generalized U-statistic and is denoted as $U_{n,s,\omega}$.
- When $N < \binom{n}{s}$, these estimators are incomplete generalized U-statistics.

Throughout this paper, we will consider generalized U-statistics as a broad class of estimators and later apply the developed results to the two-scale distributional nearest-neighbor estimator of [6]. For the purposes of this analysis, we will use the notational convention $\theta_s = \mathbb{E}\left[h_s\left(D_{[s]}\right)\right] = \mathbb{E}\left[U_{n,s,N,\omega}\left(D_{[n]}\right)\right]$. Often θ_s is assumed to be zero without loss of generality to simplify the notation, and for the purposes of the general results of this paper, we will assume the same in all of the proofs since centering the generalized U-statistic under consideration can always recover this special case. More essential, for some of the results, is the assumption that θ_s is bounded in s, a relatively benign condition that is satisfied by construction in most applications of generalized U-statistics. For the case of the TDNN estimator, we explicitly include the nonparametric regression function taking the role of θ as it is essential for the illustration of the underlying problem. To clarify the role of ω, we can conceive of it as a collection of i.i.d. random variables. That is $\omega = (W_\ell)_{\ell \in L_{n,s}}$ where each W_ℓ contains only the additional randomness injected into the kernel evaluated on its respective subsample.

Many of the most useful results in the realm of classical U-statistics are based on the celebrated Hoeffding decomposition of [7]. This technique allows us to decompose a classical U-statistic into uncorrelated components taking the form of U-statistics of order one to s, where each of these terms captures the contribution of progressively higher-order interactions of tuples of data points. In other words, each component corresponds to the projection of the kernel onto the space of functions that depend on exactly k arguments and are orthogonal to all lower-order components. It provides a powerful analytical tool for studying the variance and limiting distribution of U-statistics, especially since the components are orthogonal and, in many asymptotic settings, higher-order terms become negligible. [11] show that there is a natural extension to generalized U-statistics that includes this classical result as a special case. It stands to reason that this extended projection result will allow us to recover many classical results when applied to the generalized setting.

Definition 2.2 (Generalized Hoeffding Decomposition)**.**
Suppose $D_{[n]} = (Z_1, \ldots, Z_n)$ is a data set consisting of i.i.d. samples from F_Z and $d_{[n]}$ is

a fixed realization of the data set. Let $h_s\left(D_{[s]};\omega\right)$ be a (possibly randomized) real valued function that is permutation-symmetric in $D_{[s]}$. Let

$$h_{s|i}\left(d_{[i]}\right) = \mathbb{E}\left[h_s\left(d_{[i]}, D_{[s-i]};\omega\right)\right] - \mathbb{E}\left[h(D_{[s]};\omega)\right] \tag{2.2}$$

for $i = 1,\ldots,s$ and let

$$h_s^{(i)}\left(D_{[i]}\right) = h_{s|i}\left(D_{[i]}\right) - \sum_{j=1}^{i}\sum_{\ell\in L_{s,j}} h_s^{(j)}\left(D_\ell\right), \quad \text{for} i = 1,\ldots,s-1, \tag{2.3}$$

$$h_s^{(s)}\left(D_{[s]};\omega\right) = h_s\left(D_{[s]};\omega\right) - \sum_{j=1}^{s-1}\sum_{\ell\in L_{s,j}} h_s^{(j)}\left(D_\ell\right), \tag{2.4}$$

$$H_s^i\left(D_{[n]}\right) = \binom{n}{i}^{-1}\sum_{\ell\in L_{n,i}} h_s^{(i)}\left(D_\ell\right), \quad \text{for} i = 1,\ldots,s-1 \quad \text{and} \tag{2.5}$$

$$H_s^s\left(D_{[n]};\omega\right) = \binom{n}{s}^{-1}\sum_{\ell\in L_{n,s}} h_s^{(s)}\left(D_\ell;\omega\right). \tag{2.6}$$

The H-decomposition of a generalized complete U-statistic is expressed as

$$\begin{aligned} U_{n,s,\omega}\left(D_{[n]}\right) &= \sum_{i=1}^{s-1}\left[\binom{s}{i}\binom{n}{i}^{-1}\sum_{\ell\in L_{n,i}} h^{(i)}\left(D_\ell\right)\right] + \binom{n}{s}^{-1}\sum_{\ell\in L_{n,s}} h^{(s)}\left(D_\ell;\omega\right) \\ &= \sum_{i=1}^{s-1}\binom{s}{i}H_s^i\left(D_{[n]}\right) + H_s^s\left(D_{[n]};\omega\right). \end{aligned} \tag{2.7}$$

Furthermore, continuing to adapt the notation of [11], we define the following variance terms.

$$\zeta_{s,\omega}^c = \mathrm{Cov}\left(h\left(D_{[c]}, D_{[s-c]};\omega\right), h\left(D_{[c]}, D'_{[s-c]};\omega'\right)\right) \quad \text{for} \quad c = 1,\ldots,s-1 \tag{2.8}$$

$$\zeta_s^s = \mathrm{Cov}\left(h\left(D_{[s]};\omega\right), h\left(D_{[s]};\omega\right)\right) = \mathrm{Var}\left(h_s\left(D_{[s]};\omega\right)\right) \tag{2.9}$$

$$V_{s,\omega}^c = \mathrm{Var}\left(h_s^{(c)}\left(D_{[c]}\right)\right) \quad \text{for} \quad c = 1,\ldots,s-1 \tag{2.10}$$

$$V_s^s = \mathrm{Var}\left(h_s^{(s)}\left(D_{[s]};\omega\right)\right) \tag{2.11}$$

Here, variables with a prime (such as $D'_{[s-c]}$ or ω') denote random variables that follow the same distribution as their non-prime counterparts. They are furthermore independent of any other input variable to the U-statistic, such as each other or their non-prime counterparts. We choose to include a subscript of ω for the terms of order $1 \le c < s$ to indicate that these terms are constructed by taking expectations, including ω, forcing the additional effect of randomization to appear only in the final terms of their respective expansions. Standard results for classical U-statistics (see, for example, [8]) that can be extended to their generalized counterparts give us a number of useful results. Most importantly for our purposes is a variance

decomposition in terms of the Hoeffding projection variances.

$$\begin{aligned}\operatorname{Var}\left(U_{n,s,\omega}\left(D_{[n]}\right)\right) &= \sum_{j=1}^{s-1}\binom{s}{j}^2 \operatorname{Var}\left(H_s^j\left(D_{[n]}\right)\right) + \operatorname{Var}\left(H_s^s\left(D_{[n]};\omega\right)\right) \\ &= \sum_{j=1}^{s-1}\binom{s}{j}^2\binom{n}{j}^{-1} \operatorname{Var}\left(h_s^{(j)}\left(D_{[j]}\right)\right) + \binom{n}{s}^{-1} \operatorname{Var}\left(h_s^{(s)}\left(D_{[s]};\omega\right)\right) \quad (2.12) \\ &= \sum_{j=1}^{s-1}\binom{s}{j}^2\binom{n}{j}^{-1} V_{s,\omega}^j + \binom{n}{s}^{-1} V_s^s\end{aligned}$$

Clearly, this result contains the Hoeffding decomposition for classical U-statistics as a special case, namely for a fixed s and absent additional randomization in the form of ω. Thus, it also recovers the associated variance decomposition for classical U-statistics in a natural fashion.

3. Consistent Variance Estimation

Theorem 1 and Theorem 2 of [11] establish the normality of generalized U-statistics under suitable dominance conditions for the variance of the Hájek projection. As the exact same condition will be central to the consistency of the jackknife variance estimators considered in this paper, we restate it by itself.

Assumption 1 (Asymptotic Hájek Dominance Condition)**.**
Consider a potentially incomplete generalized U-statistic $U_{n,s,N,\omega}$. If

$$\frac{s}{n}\left(\frac{\zeta_s^s}{s\zeta_{s,\omega}^1} - 1\right) \longrightarrow 0, \quad (3.1)$$

we say $U_{n,s,N,\omega}$ satisfies the asymptotic Hájek dominance condition.

As $n \to \infty$ and $s = s(n) \to \infty$, both the full-kernel and first-order projection variances vanish, and this condition provides an analog of well-known degeneracy conditions in the classical U-statistic literature without relying on exact zeroes to occur. This condition ensures that the variance of the Hájek projection asymptotically dominates the variance of the complete generalized U-statistic.

Lemma 3.1 (Dominance of Hájek Projection Variance)**.**
Let $\mathrm{U}_{n,s,\omega}\left(D_{[n]}\right)$ *be a complete generalized U-statistic. Let the kernel variance terms* ζ_s^s *and* $\zeta_{s,\omega}^1$ *be defined as in Equations 2.8 and 2.9. Then under Assumption 1, the Hájek projection term asymptotically dominates the variance of the generalized U-statistic*

$$\frac{n}{s^2}\frac{Var\left(\mathrm{U}_{n,s,\omega}\left(D_{[n]}\right)\right)}{\zeta_{s,\omega}^1} \longrightarrow 1. \quad (3.2)$$

Thus, informally, Assumption 1 implies that the bulk of the U-statistic's variance is explained by the contributions of individual observations, rather than their (higher-order) interactions. Next, we introduce a condition from [12] that becomes relevant when we are considering the incomplete evaluation of generalized U-statistics. It will ensure that asymptotically we will consider a sufficiently large number of samples in each U-statistic that is evaluated as part of the jackknife to achieve our goal of consistent variance estimation.

Assumption 2 (Asymptotically-Sufficient Sampling Condition)**.**
Consider a potentially incomplete generalized U-statistic $U_{n,s,N,\omega}$. If

$$\frac{n}{Ns\zeta^1_{s,\omega}} \longrightarrow 0, \tag{3.3}$$

we say $U_{n,s,N,\omega}$ satisfies the asymptotically-sufficient sampling condition.

If $\zeta^1_{s,\omega}$ is bounded, this condition simplifies to $\frac{n}{Ns} \to 0$, in essence enforcing that the number of observations used for the evaluation of the generalized U-statistic, counted including multiples across overlapping subsamples, grows faster than the sample size. Thus, in expectation, the number of occurrences of each individual observation diverges as the sample size increases. This can, for example, occur in the bounded setting when $N = n^{1+\delta}$ and $s \to \infty$ with $s = o(n)$ and $\delta > 0$. The core contribution of this paper is then to justify the use of the jackknife for variance estimation for generalized U-statistics that satisfy said dominance condition. To this end, we start by defining the object for which we aim to achieve ratio-consistency,

$$\sigma^2_n = \mathrm{Var}\left(U_{n,s,N,\omega}\left(D_{[n]}\right)\right). \tag{3.4}$$

To indicate the dependence of this variance on n (and with it s and N), we explicitly include a subscript. The variance estimators that we consider as part of this paper are as follows.

Definition 3.2 (Jackknife Variance Estimators)**.**
We consider the *Jackknife Variance Estimator*

$$\widehat{\sigma}^2_{JK}\left(D_{[n]};\omega\right) \coloneqq \frac{n-1}{n}\sum_{i=1}^{n}\left(U_{n,s,N,\omega}\left(D_{[n],-i}\right) - U_{n,s,N,\omega}\left(D_{[n]}\right)\right)^2 \tag{3.5}$$

and the *delete-d Jackknife Variance Estimator*

$$\widehat{\sigma}^2_{JKD}\left(D_{[n]};d,\omega\right) \coloneqq \frac{n-d}{d}\binom{n}{d}^{-1}\sum_{\ell\in L_{n,d}}\left(U_{n,s,N,\omega}\left(D_{[n],-\ell}\right) - U_{n,s,N,\omega}\left(D_{[n]}\right)\right)^2. \tag{3.6}$$

We first consider complete generalized U-statistics and point out that analogous results on the consistency of the pseudo-infinitesimal jackknife estimator are derived in [12].

Theorem 3.3 (Variance Estimation for Complete Generalized U-Statistics)**.**
Let $U_{n,s,\omega}$ be a complete generalized U-statistic with $s = o(n)$ satisfying Assumption 1. Let $\widehat{\sigma}^2_{JKD}\left(D_{[n]};d,\omega\right)$ be the associated delete-d jackknife variance estimator as defined in Equation 3.6 such that $sd = o(n)$, then

$$\frac{\widehat{\sigma}^2_{JKD}\left(D_{[n]};d,\omega\right)}{\sigma^2_n} \longrightarrow_p 1. \tag{3.7}$$

Thus, as a special case for $d = 1$ we find that

$$\frac{\widehat{\sigma}^2_{JK}\left(D_{[n]};\omega\right)}{\sigma^2_n} \longrightarrow_p 1. \tag{3.8}$$

Remark 1. Even though $sd = o(n)$ allows for d to grow as the sample size increases, in practice the most relevant scenario will be arbitrary fixed-d regimes. The consistency result then allows us to employ the well-known idea of combining delete-d jackknife variance estimates corresponding to different fixed values of d to eliminate lower-order bias terms. While beyond the scope of this paper, this could potentially improve the performance of variance estimates in finite samples. In contrast, for a classical U-statistic with fixed s this result justifies the use of delete-d Jackknife variance estimators for the regime $d = o(n)$.

Of high practical relevance is the extension to potentially incomplete generalized U-statistics. This is due to the fact that in the absence of closed-form solutions for the evaluation of a generalized U-statistic, the evaluation of the kernel on all subsets of cardinality s is often prohibitive. Instead, we often rely on a Bernoulli sampling scheme as introduced in Definition 2.1. Thus, a strictly stronger result such as the following is of great importance.

Theorem 3.4 (Variance Estimation for Incomplete Generalized U-Statistics)**.**
Let $U_{n,s,N,\omega}$ be a potentially incomplete generalized U-statistic with $s = o(n)$ satisfying the asymptotic Hájek dominance condition (Assumption 1) and the asymptotically-sufficient sampling condition (Assumption 2). Furthermore, let θ and ζ_s^s be bounded in s. Let $\widehat{\sigma}^2_{JKD}\left(D_{[n]}; d, \omega\right)$ be the associated delete-d jackknife variance estimator as defined in Equation 3.6 with $sd = o(n)$, then

$$\frac{\widehat{\sigma}^2_{JKD}\left(D_{[n]}; d, \omega\right)}{\sigma_n^2} \longrightarrow_p 1. \tag{3.9}$$

Thus, as a special case we find that

$$\frac{\widehat{\sigma}^2_{JK}\left(D_{[n]}; \omega\right)}{\sigma_n^2} \longrightarrow_p 1. \tag{3.10}$$

4. Application to Two-Scale Distributional Nearest Neighbor Estimator

One recent example of an estimator covered by these results is the two-scale distributional nearest-neighbor regression estimator of [6]. Its purpose is nonparametric regression, which we will consider using the following setup.

Assumption 3 (Nonparametric Regression DGP)**.**
The observed data consist of an i.i.d. sample taking the following form.

$$D_{[n]} = \{Z_i = (X_i, Y_i)\}_{i=1}^n \quad \text{from the model} \quad Y = \mu(X) + \varepsilon, \tag{4.1}$$

where $Y \in \mathcal{Y} \subset \mathbb{R}$ is the response, $X \in \mathcal{X} \subset \mathbb{R}^k$ is a feature vector of fixed dimension k distributed according to a density function f with associated probability measure φ on $\mathcal{X}$, and $\mu(x)$ is the unknown mean regression function. ε is the unobservable model error on which we impose the following conditions.

$$\mathbb{E}\left[\varepsilon | X\right] = 0, \quad \text{Var}\left(\varepsilon | X = x\right) = \sigma_\varepsilon^2\left(x\right) \tag{4.2}$$

Let the distribution induced by this model be denoted by P and thus $Z_i = (X_i, Y_i) \overset{\text{iid}}{\sim} P$.

The results from [6] cover this slightly extended heteroskedastic setup under the assumptions presented in the original paper, with only minor adjustments to the proofs. Throughout the treatment of the TDNN estimator, we will additionally rely on a number of assumptions that are more technical in nature.

Assumption 4 (Technical Assumptions)**.**
The following conditions hold.

- The feature space $\mathcal{X} = \text{supp}(X)$ is a bounded, compact subset of $\mathbb{R}^k$
- The density $f(\cdot)$ is bounded away from 0 and ∞, i.e., $\forall x \in \mathcal{X}: \quad 0 < \underline{\mathfrak{f}} \leq f(x) \leq \overline{\mathfrak{f}} < \infty$.
- $f(\cdot)$ and $\mu(\cdot)$ are four times continuously differentiable with bounded second, third, and fourth-order partial derivatives. Specifically, $\forall x \in \mathcal{X} \quad \forall (i,j,l,m) \in [k]^4$:

$$\begin{aligned} -\infty \ < \ \underline{\mathfrak{f}}' \ \leq \partial_{i,j} f(x), \partial_{i,j,m} f(x), \partial_{i,j,l,m} f(x) \ \leq \ \overline{\mathfrak{f}}' \ < \infty \\ -\infty \ < \underline{\mathfrak{m}}' \ \leq \partial_{i,j} \mu(x), \partial_{i,j,m} \mu(x), \partial_{i,j,l,m} \mu(x) \ \leq \overline{\mathfrak{m}}' \ < \infty \end{aligned}$$

- $\mu(\cdot) \in L^2(\mathcal{X})$ is a square-integrable function on $\mathcal{X}$ with respect to φ.
- $\sigma_\varepsilon^2 : \mathcal{X} \longrightarrow \mathbb{R}_{>0}$ is a continuous function on $\mathcal{X}$ that is square-integrable with respect to φ. Thus, the variance is bounded above by some $\overline{\sigma}_\varepsilon^2 > 0$.

Remark 2. There is considerable potential to relax these assumptions at the cost of requiring both less interpretable conditions and more technically sophisticated proofs. For example, the bounded derivatives condition can be relaxed to hold only in a neighborhood of x while requiring a weaker, more complex condition on the behavior of the derivatives beyond that neighborhood.

The TDNN estimator is an extension of a U-statistic type estimator developed and analyzed in [14] and [3]. This original estimator, that we will refer to as the distributional nearest-neighbor (DNN) regression estimator, is constructed in the following way. Given a fixed feature vector of interest x, we first order the sample based on the distance to the point of interest.

$$||X_{(1)} - x||_2 < ||X_{(2)} - x||_2 < \ldots < ||X_{(n)} - x||_2 \tag{4.3}$$

Due to our assumption that X is continuously distributed, ties occur with probability zero, and thus we assign the same rank to tied observations for notational simplicity. Let $\text{rk}(x; X, D)$ denote the rank relative to a point of interest x that would be assigned to an observation with covariate vector X if it was added to a sample D. This enables us to define a data-driven function κ in the following way.

$$\kappa(x; Z_i, D_\ell) = \mathbb{1}\left(\text{rk}(x; X_i, D_\ell) = 1\right) \tag{4.4}$$

Defining the kernel function, $h_s(x; D_\ell) \coloneqq \sum_{i=1}^n \mathbb{1}(i \in \ell)\kappa(x; Z_i, D_\ell) Y_i$, the DNN estimator is given by the following U-statistic.

$$\tilde{\mu}_s(x; D_{[n]}) = \binom{n}{s}^{-1} \sum_{\ell \in L_{n,s}} h_s(x; D_\ell) \tag{4.5}$$

What makes the TDNN estimator special is the way it combines two subsampling scales $1 \leq s_1 < s_2 \leq n$ to eliminate the first-order bias, similar to higher-order kernels in the theory

on nonparametric kernel regression. We will denote the vector (s_1, s_2) by $\mathfrak{S}$ to simplify the notation that occurs in many of the proofs. Specifically, we will subscript a number of variables by $\mathfrak{S}$ to signify that they are associated with a sequence of TDNN estimators as n increases. Given the two subsampling scales s_1 and s_2, we obtain the two corresponding weights.

$$w_1^* = \frac{1}{1 - (s_1/s_2)^{-2/k}} \quad \text{and} \quad w_2^* = 1 - w_1^*(s_1, s_2) \tag{4.6}$$

Using these weights, [6] define the corresponding TDNN estimator as follows.

$$\widehat{\mu}_{\mathfrak{S}}\left(x; D_{[n]}\right) = w_1^* \tilde{\mu}_{s_1}\left(x; D_{[n]}\right) + w_2^* \tilde{\mu}_{s_2}\left(x; D_{[n]}\right) = \binom{n}{s_2}^{-1} \sum_{\ell \in L_{n,s_2}} h_{\mathfrak{S}}(x; D_\ell) \tag{4.7}$$

Here, the kernel of the TDNN estimator takes the following form.

$$\begin{aligned} h_{\mathfrak{S}}\left(x; D_{[s_2]}\right) &= w_1^* \left[\binom{s_2}{s_1}^{-1} \sum_{\ell \in L_{s_2,s_1}} h_{s_1}\left(x; D_\ell\right) \right] + w_2^* h_{s_2}\left(x; D_{[s_2]}\right) \\ &= w_1^* \tilde{\mu}_{s_1}\left(x; D_{[s_2]}\right) + w_2^* h_{s_2}\left(x; D_{[s_2]}\right) \end{aligned} \tag{4.8}$$

A condition that ensures that the TDNN estimator does not become essentially equivalent to a DNN estimator restricts the ratio of the kernel orders. This assumption is thus more of a guardrail to ensure that the estimator does not degenerate to the point where the two-scale component becomes effectively redundant.

Assumption 5 (Bounded Ratio of Kernel-Orders)**.**
There is a constant $\mathfrak{c} \in (0, 1/2)$ such that the ratio of kernel orders is bounded in the following way.

$$\forall n: \quad 0 < \mathfrak{c} \leq s_1/s_2 \leq 1 - \mathfrak{c} < 1. \tag{4.9}$$

In the original paper, consistency of the jackknife variance estimator is established under the strong condition that $s_2 = o(n^{1/3})$. This condition, which restricts its use in practice by narrowing the choice set of kernel orders, can be loosened considerably using the results developed in this paper. Specifically, we achieve consistent jackknife-based variance estimation in the regime $s_2 = o(n)$, allowing for potentially superior bias-variance trade-offs. However, in contrast to the generalized U-statistics case, the variance term of interest is now localized. Thus, we use the following notation.

$$\sigma_n^2(x) = \text{Var}\left(\widehat{\mu}_{\mathfrak{S}}\left(x; D_{[n]}\right)\right) \tag{4.10}$$

Similarly, we will include x as an argument in the jackknife variance estimators to make the localized nature of the estimator explicit.

Theorem 4.1 (The TDNN Estimator satisfies the Asymptotic Hájek Dominance Condition)**.**
Consider a data-generating process as outlined in Assumption 3 and Assumption 4. Let $0 < s_1 < s_2 = o(n)$ *be such that Assumption 5 holds. Then, the DNN and TDNN regression estimators satisfy the Asymptotic Hájek Dominance Condition.*

Clearly, this result enables us to obtain ratio-consistent variance estimates for the TDNN estimator in a wider set of regimes than considered in [6]. As a side note, it is worth pointing out that a number of conditions we assume in the statement of this theorem are not necessarily connected to the ratio consistency of the variance estimator directly. Instead, several of them play crucial roles in the derivation of asymptotic normality and the elimination of the first-order bias in [6]. As the authors of the original paper point out, a number of their assumptions are in place to simplify the presentation of results and can be replaced by weaker conditions at the cost of weaker interpretability. This logic fully carries over to our paper, where we do not seek to generalize the original results beyond our improved conditions for variance estimation. This new result in turn brings the variance estimation procedures in line with the authors' asymptotic normality results. Specifically, we obtain the following result.

Theorem 4.2 (Ratio-Consistent Variance Estimation for the TDNN Estimator)**.**
Let $0 < s_1 < s_2 = o(n)$ *and* $s_2 d = o(n)$. *Then the following ratio consistency holds for the TDNN regression estimator and its associated jackknife variance estimators given the data-generating process outlined in Assumption 3 and Assumption 4, and kernel orders satisfying Assumption 5. Then,*

$$\frac{\widehat{\sigma}^2_{JKD}\left(x, D_{[n]}; d\right)}{\sigma^2_n\left(x\right)} \longrightarrow_p 1. \tag{4.11}$$

Thus, as a special case we find that

$$\frac{\widehat{\sigma}^2_{JK}\left(x, D_{[n]}\right)}{\sigma^2_n\left(x\right)} \longrightarrow_p 1. \tag{4.12}$$

As the DNN and TDNN regression estimators allow for convenient closed-form representations, we refrain from considering the incomplete cases. The main motivation for the derivation of these results is clearly asymptotically valid inference. Thus, as a simple corollary, we can observe the following.

Theorem 4.3 (Asymptotically Valid Inference with the TDNN Estimator)**.**
Let $0 < s_1 < s_2 = o(n)$ *and* $s_2 d = o(n)$. *Then the following holds for the TDNN regression estimator and its associated jackknife variance estimators given a data generating process as outlined in Assumption 3 and Assumption 4, and kernel orders satisfying Assumption 5.*

$$\frac{\widehat{\mu}_{\mathfrak{S}}\left(x; D_{[n]}\right) - \mu\left(x\right)}{\sqrt{\widehat{\sigma}^2_{JKD}\left(x, D_{[n]}; d\right)}} \rightsquigarrow \mathcal{N}\left(0, 1\right). \tag{4.13}$$

Thus, as a special case we find that

$$\frac{\widehat{\mu}_{\mathfrak{S}}\left(x; D_{[n]}\right) - \mu\left(x\right)}{\sqrt{\widehat{\sigma}^2_{JK}\left(x, D_{[n]}\right)}} \rightsquigarrow \mathcal{N}\left(0, 1\right). \tag{4.14}$$

Thus, in the Hájek projection dominance regime, jackknife-based studentization yields asymptotically valid inference for the TDNN estimator without requiring further modifications.

Remark 3. It is worth noting that several other base learners have already been investigated by [11] for whether they satisfy the asymptotic Hájek dominance condition. This includes a number of classical U-statistics such as the mean and sample variance, giving an alternative perspective on the consistency of the jackknife in these well-understood scenarios. Furthermore, the authors investigate estimators such as the classical k-nearest neighbors estimator (kNN), the k-potential nearest neighbors estimator (kPNN), and a number of tree-based learners such as honest and double-sampling trees. This in turn extends the immediate applicability of the results developed in the paper at hand to other, potentially more popular generalized U-statistics without requiring additional arguments.

5. Conclusion and Outlook

This paper establishes sufficient conditions under which the nonparametric Jackknife yields consistent variance estimates for generalized U-statistics. The key insight is that consistency reduces to a simple and interpretable criterion: asymptotic dominance of the Hájek projection variance. This condition, which requires that the variance of the full statistic be asymptotically governed by its first-order projection, provides a unified foundation for understanding variance estimation across a broad class of randomized, incomplete, and infinite-order estimators. Our results clarify the relationship between the classical Jackknife and its delete-d variants, and show that consistency can be achieved without modifying the estimators themselves or imposing strong moment or smoothness conditions. The theory applies naturally to modern machine learning-inspired procedures, such as distributional nearest-neighbor methods. As an illustrative example, we showed that the Two-Scale Distributional Nearest-Neighbor estimator of [6] satisfies the proposed condition under a moderately fast-growing kernel order. Looking ahead, similar arguments can be made for generalized U-statistics that exhibit degeneracy, where the appropriate dominance condition involves the first nonzero variance projection term. The structure of the proofs given here, along with closely related ideas in [12] concerning the infinitesimal Jackknife, suggests that this extension is primarily notational, as the underlying argument translates directly to degenerate cases. This would effectively generalize many of the classical results of [2] to a broad and practically relevant class of estimators.

Beyond the degenerate case, an important direction for future work is the application of these variance bounds to plug-in estimators involving generalized U-statistics in the second stage. The projection dominance perspective may serve as a modular tool for deriving valid inference in such estimators, including those used for conditional average treatment effect estimation. In this setting, using a distributional nearest-neighbor kernel in the second stage allows observations to be weighted according to a localized criterion tailored to the target covariate value. This construction may lead to a novel class of doubly robust estimators, particularly well-suited to high-dimensional regimes where standard kernel-based weighting schemes tend to fail. Another natural direction for future work lies in the analysis of random forest-type estimators. Many random forest variants, including honest regression forests and distributional forests, can be expressed as incomplete or randomized generalized U-statistics with complex dependence structures. In such cases, the variance of the estimator is often driven by the interplay between the sampling scheme and the localized weighting induced by the tree structure. The asymptotic Hájek projection dominance condition introduced in this paper offers a tractable criterion for diagnosing when variance estimation via the Jackknife

is valid. A deeper understanding of projection dominance in random forests may help clarify the asymptotic behavior of forest-based plug-in estimators, guide the design of subsampling schemes that ensure valid inference, and provide new insight into the limits of Jackknife and bootstrap procedures in high-dimensional or adaptively regularized models.

Appendix A: Proofs for Generalized U-Statistics Results

To show the consistency of the jackknife variance estimators under consideration, we will in part rely on the following basic result from [12].

Lemma A.1 ([12] - Lemma C.1.)**.**
Suppose that $\sum X_i^2 \xrightarrow{p} 1, \sum \mathbb{E}\left[X_i^2\right] \longrightarrow 1$, *and* $\sum_{i=1}^n \mathbb{E}\left[Y_i^2\right] \longrightarrow 0$, *then*

$$\sum [X_i + Y_i]^2 \xrightarrow{p} 1 \quad \textit{and} \quad \mathbb{E}\left[\sum (X_i + Y_i)^2\right] \longrightarrow 1. \tag{A.1}$$

Proof of Lemma 3.1.

$$\begin{aligned} 1 \leq \frac{n}{s^2} \frac{\operatorname{Var}\left(\mathrm{U}_{n,s,\omega}\left(D_{[n]}\right)\right)}{\zeta_{s,\omega}^1} &= \left(\frac{s^2}{n}\zeta_{s,\omega}^1\right)^{-1} \left(\sum_{j=1}^{s-1} \binom{s}{j}^2 \binom{n}{j}^{-1} V_{s,\omega}^j + \binom{n}{s}^{-1} V_s^s\right) \\ &\leq 1 + \left(\frac{s^2}{n}\zeta_{s,\omega}^1\right)^{-1} \frac{s^2}{n^2} \left(\sum_{j=2}^{s} \binom{s}{j} V_{s,\omega}^j + V_s^s\right) \\ &\leq 1 + \frac{s}{n}\left(\frac{\zeta_s^s}{s\zeta_{s,\omega}^1} - 1\right) \longrightarrow 1. \end{aligned} \tag{A.2}$$

□

Proof of Theorem 3.3.
See next page.

We start by making use of the Hoeffding decomposition in the following way.

$$\begin{aligned}
\widehat{\sigma}^2_{JKD}\left(D_{[n]};d\right) &= \frac{n-d}{d}\binom{n}{d}^{-1}\sum_{\ell\in L_{n,d}}\left[\sum_{j=1}^{s-1}\binom{s}{j}H^j_{s,\omega}\left(D_{[n]}\right)-H^j_{s,\omega}\left(D_{[n],-\ell}\right)d+H^s_s\left(D_{[n]}\right)-H^s_s\left(D_{[n],-\ell}\right)\right]^2 \\
&= \frac{n-d}{d}\binom{n}{d}^{-1}\sum_{\ell\in L_{n,d}}\left[\sum_{j=1}^{s-1}\binom{s}{j}\left(\binom{n}{j}^{-1}\sum_{\iota\in L_{n,j}}h^{(j)}_{s,\omega}(D_\iota)-\binom{n-d}{j}^{-1}\sum_{\iota\in L_j([n]\setminus\ell)}h^{(j)}_{s,\omega}(D_\iota)\right)\right. \\
&\quad \left.+\left(\binom{n}{s}^{-1}\sum_{\iota\in L_{n,s}}h^{(s)}_s(D_\iota)-\binom{n-d}{s}^{-1}\sum_{\iota\in L_s([n]\setminus\ell)}h^{(s)}_s(D_\iota)\right)\right]^2
\end{aligned} \tag{A.3}$$

We split this expression into a Hájek term and a residual as the asymptotic Hájek dominance condition will ensure that the expression of interest will be dominated by the former.

$$\begin{aligned}
&\widehat{\sigma}^2_{JKD}\left(D_{[n]};d\right) = \frac{n-d}{d}\binom{n}{d}^{-1}\sum_{\ell\in L_{n,d}}\left[\frac{s}{n}\sum_{i\in\ell}h^{(1)}_{s,\omega}(Z_i)+\sum_{i\in[n]\setminus\ell}\left(\frac{s}{n}-\frac{s}{n-d}\right)h^{(1)}_{s,\omega}(Z_i)\right. \\
&\quad \left.+\sum_{j=2}^{s-1}\binom{j}{s}\left(\binom{n}{j}^{-1}\sum_{\iota\in L_{n,j}}h^{(j)}_{s,\omega}(D_\iota)-\binom{n-d}{j}^{-1}\sum_{\iota\in L_j([n]\setminus\ell)}h^{(j)}_{s,\omega}(D_\iota)\right)+\binom{n}{s}^{-1}\sum_{\iota\in L_{n,s}}h^{(s)}_s(D_\iota)-\binom{n-d}{s}^{-1}\sum_{\iota\in L_s([n]\setminus\ell)}h^{(s)}_s(D_\iota)\right]^2 \\
&= \frac{n-d}{d}\binom{n}{d}^{-1}\left(\frac{s}{n}\right)^2\sum_{\ell\in L_{n,d}}\left[\sum_{i\in\ell}h^{(1)}_{s,\omega}(Z_i)-\frac{d}{n-d}\sum_{i\in[n]\setminus\ell}h^{(1)}_{s,\omega}(Z_i)\right. \\
&\quad \left.+\frac{n}{s}\sum_{j=2}^{s-1}\binom{s}{j}\left(\binom{n}{j}^{-1}\sum_{\iota\in L_{n,j}}h^{(j)}_{s,\omega}(D_\iota)-\binom{n-d}{j}^{-1}\sum_{\iota\in L_j([n]\setminus\ell)}h^{(j)}_{s,\omega}(D_\iota)\right)+\binom{n}{s}^{-1}\sum_{\iota\in L_{n,s}}h^{(s)}_s(D_\iota)-\binom{n-d}{s}^{-1}\sum_{\iota\in L_s([n]\setminus\ell)}h^{(s)}_s(D_\iota)\right]^2 \\
&=: \frac{s^2}{(n-d)\cdot n^2}\binom{n}{d}^{-1}\sum_{\ell\in L_{n,d}}\left[\frac{1}{\sqrt{d}}\sum_{i\in\ell}h^{(1)}_{s,\omega}(Z_i)+T_\ell\right]^2
\end{aligned} \tag{A.4}$$

Next, we want to show that $\sum_{i\in\ell} h^{(1)}_{s,\omega}(Z_i)$ dominates T_ℓ in the sense of Lemma A.1. Since Lemma A.1 does not depend on any particular independence assumptions of summands etc. this is a relatively straightforward adaptation. Thus, consider the following for an arbitrary fixed index-subset ℓ with cardinality d.

$$\begin{aligned}\mathbb{E}\left[\left(\frac{1}{\sqrt{d}}\sum_{i\in\ell} h^{(1)}_{s,\omega}(Z_i)\right)^2\right] &= \frac{1}{d}\mathbb{E}\left[\sum_{i\in\ell}\sum_{j\in\ell} h^{(1)}_{s,\omega}(Z_i)h^{(1)}_{s,\omega}(Z_j)\right]\\ &= \frac{1}{d}\sum_{i\in\ell}\sum_{j\in\ell}\mathbb{E}\left[h^{(1)}_{s,\omega}(Z_i)h^{(1)}_{s,\omega}(Z_j)\right]\\ &= \frac{|\ell|}{d}\cdot\mathbb{E}\left[\left(h^{(1)}_{s,\omega}(Z_1)\right)^2\right] = \zeta^1_{s,\omega}\end{aligned} \tag{A.5}$$

For the error term we introduce a case distinction. Case one corresponds to parameter choices where $s \geq d$ and thus takes the following form.

$$\begin{aligned}T_\ell = \frac{\sqrt{d}}{n-d}\sum_{i\in[n]\setminus\ell} h^{(1)}_{s,\omega}(Z_i) + \frac{n}{s\sqrt{d}}\Bigg\{&\sum_{j=2}^{d}\binom{s}{j}\Bigg[\binom{n}{j}^{-1}\sum_{a=1}^{j}\sum_{\substack{\varkappa\in L_a(\ell)\\ \varrho\in L_{j-a}([n]\setminus\ell)}} h^{(j)}_{s,\omega}(D_{\varkappa\cup\varrho})\\ &+\left(\binom{n}{j}^{-1}-\binom{n-d}{j}^{-1}\right)\sum_{\iota\in L_j([n]\setminus\ell)} h^{(j)}_{s,\omega}(D_\iota)\Bigg]\\ +\sum_{j=d+1}^{s-1}\binom{s}{j}&\Bigg[\binom{n}{j}^{-1}\sum_{a=1}^{d}\sum_{\substack{\varkappa\in L_a(\ell)\\ \varrho\in L_{j-a}([n]\setminus\ell)}} h^{(j)}_{s,\omega}(D_{\varkappa\cup\varrho})\\ &+\left(\binom{n}{j}^{-1}-\binom{n-d}{j}^{-1}\right)\sum_{\iota\in L_j([n]\setminus\ell)} h^{(j)}_{s,\omega}(D_\iota)\Bigg]\\ +\binom{n}{s}^{-1}\sum_{a=1}^{d}&\sum_{\substack{\varkappa\in L_a(\ell)\\ \varrho\in L_{s-a}([n]\setminus\ell)}} h^{(s)}_{s}(D_{\varkappa\cup\varrho}) + \left(\binom{n}{s}^{-1}-\binom{n-d}{s}^{-1}\right)\sum_{\iota\in L_s([n]\setminus\ell)} h^{(s)}_{s}(D_\iota)\Bigg\}\end{aligned} \tag{A.6}$$

Case two covers setups of the form $s < d$ and thus takes the following form. Note that we distinguish this second case from the first by using $\star$ as a superscript.

$$\begin{aligned}T^\star_\ell = \frac{\sqrt{d}}{n-d}\sum_{i\in[n]\setminus\ell} h^{(1)}_{s,\omega}(Z_i) + \frac{n}{s\sqrt{d}}\Bigg\{&\sum_{j=2}^{s-1}\binom{s}{j}\Bigg(\binom{n}{j}^{-1}\sum_{a=1}^{j}\sum_{\substack{\varkappa\in L_a(\ell)\\ \varrho\in L_{j-a}([n]\setminus\ell)}} h^{(j)}_{s,\omega}(D_{\varkappa\cup\varrho}) + \left(\binom{n}{j}^{-1}-\binom{n-d}{j}^{-1}\right)\sum_{\iota\in L_j([n]\setminus\ell)} h^{(j)}_{s,\omega}(D_\iota)\Bigg)\\ +\binom{n}{s}^{-1}\sum_{a=1}^{s}&\sum_{\substack{\varkappa\in L_a(\ell)\\ \varrho\in L_{s-a}([n]\setminus\ell)}} h^{(s)}_{s}(D_{\varkappa\cup\varrho}) + \left(\binom{n}{s}^{-1}-\binom{n-d}{s}^{-1}\right)\sum_{\iota\in L_s([n]\setminus\ell)} h^{(s)}_{s}(D_\iota)\Bigg\}\end{aligned} \tag{A.7}$$

Now, recall the Chu-Vandermonde identity, i.e. for any nonnegative integers r, m, n the following holds.

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k} \tag{A.8}$$

Having separated these two cases, we continue by investigating the expectation of their respective squares. Beginning with case one ($s \geq d$), we find the following.

$$\begin{aligned}
\mathbb{E}\left[(T_\ell)^2\right] &= \frac{d}{n-d}V^1_{s,\omega} + \frac{n^2}{s^2 d}\Bigg\{ \\
&\sum_{j=2}^{d}\binom{s}{j}^2\left(\binom{n}{j}^{-2}\sum_{a=1}^{j}\binom{d}{a}\binom{n-d}{j-a} + \left[\binom{n}{j}^{-1} - \binom{n-d}{j}^{-1}\right]^2\binom{n-d}{j}\right)V^j_{s,\omega} \\
&\quad + \sum_{j=d+1}^{s-1}\binom{s}{j}^2\left(\binom{n}{j}^{-2}\sum_{a=1}^{d}\binom{d}{a}\binom{n-d}{j-a} + \left[\binom{n}{j}^{-1} - \binom{n-d}{j}^{-1}\right]^2\binom{n-d}{j}\right)V^j_{s,\omega} \\
&\quad + \left(\binom{n}{s}^{-2}\sum_{a=1}^{d}\binom{d}{a}\binom{n-d}{s-a} + \left[\binom{n}{s}^{-1} - \binom{n-d}{s}^{-1}\right]^2\binom{n-d}{s}\right)V^s_s\Bigg\} \\
&\overset{(\star)}{=} \frac{d}{n-d}V^1_{s,\omega} + \frac{n}{sd}\Bigg\{\sum_{j=2}^{d}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}}\frac{\binom{s}{j}}{\binom{n}{j}}\left(\binom{n}{j} - \binom{n-d}{j} + \left[1 - \binom{n}{j}\binom{n-d}{j}^{-1}\right]^2\binom{n-d}{j}\right)V^j_{s,\omega} \\
&\quad + \sum_{j=d+1}^{s-1}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}}\frac{\binom{n-d}{j}}{\binom{n}{j}}\left(\sum_{a=1}^{d}\frac{\binom{d}{a}\binom{n-d}{j-a}}{\binom{n-d}{j}} + \left[1 - \binom{n}{j}\binom{n-d}{j}^{-1}\right]^2\right)\binom{s}{j}V^j_{s,\omega} \\
&\quad + \frac{\binom{n-d}{s}}{\binom{n-1}{s-1}\binom{n}{s}}\left(\sum_{a=1}^{d}\frac{\binom{d}{a}\binom{n-d}{s-a}}{\binom{n-d}{s}} + \left[1 - \binom{n}{s}\binom{n-d}{s}^{-1}\right]^2\right)V^s_s\Bigg\}
\end{aligned} \tag{A.9}$$

The equality marked by $(\star)$ holds by the Chu-Vandermonde identity - specifically with respect to the equivalent expression for the sum in the second term.

Similarly, for the second case ($s < d$), we can make the following observation.

$$\begin{aligned}
\mathbb{E}\left[(T_\ell^\star)^2\right] &= \frac{d}{n-d}V_{s,\omega}^1 + \frac{n^2}{s^2 d}\Bigg\{ \\
&\sum_{j=2}^{s-1}\binom{s}{j}^2\left(\binom{n}{j}^{-2}\sum_{a=1}^{j}\binom{d}{a}\binom{n-d}{j-a} + \left[\binom{n}{j}^{-1} - \binom{n-d}{j}^{-1}\right]^2\binom{n-d}{j}\right)V_{s,\omega}^j \\
&\quad + \left(\binom{n}{s}^{-2}\sum_{a=1}^{s}\binom{d}{a}\binom{n-d}{s-a} + \left[\binom{n}{s}^{-1} - \binom{n-d}{s}^{-1}\right]^2\binom{n-d}{s}\right)V_s^s\Bigg\} \\
&\overset{(\circ)}{=} \frac{d}{n-d}V_{s,\omega}^1 + \frac{n}{sd}\Bigg\{\sum_{j=2}^{s-1}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}}\frac{\binom{s}{j}}{\binom{n}{j}}\left(\binom{n}{j} - \binom{n-d}{j} + \left[1-\binom{n}{j}\binom{n-d}{j}^{-1}\right]^2\binom{n-d}{j}\right)V_{s,\omega}^j \\
&\quad + \binom{n-1}{s-1}^{-1}\binom{n}{s}^{-1}\left(\binom{n}{s} - \binom{n-d}{s} + \left[1-\binom{n}{s}\binom{n-d}{s}^{-1}\right]^2\binom{n-d}{s}\right)V_s^s\Bigg\}
\end{aligned} \tag{A.10}$$

Here, as before, the equality marked by $(\circ)$ holds by the Chu-Vandermonde identity.

Continuing the analysis, we find the following.

$$\begin{aligned}
\mathbb{E}\left[(T_\ell)^2\right] &= \frac{d}{n-d}V_{s,\omega}^1 \\
&\quad + \frac{n}{sd}\Bigg\{\sum_{j=2}^{d}\frac{\binom{s-1}{j-1}\binom{n-d}{j}}{\binom{n-1}{j-1}\binom{n}{j}}\left(\binom{n}{j}\binom{n-d}{j}^{-1} - 1 + \left[1-\binom{n}{j}\binom{n-d}{j}^{-1}\right]^2\right)\binom{s}{j}V_{s,\omega}^j \\
&\quad + \sum_{j=d+1}^{s-1}\frac{\binom{s-1}{j-1}\binom{n-d}{j}}{\binom{n-1}{j-1}\binom{n}{j}}\left(\frac{\binom{n}{j}}{\binom{n-d}{j}}\sum_{a=1}^{d}\frac{\binom{d}{a}\binom{n-d}{j-a}}{\binom{n}{j}} + \left[1-\binom{n}{j}\binom{n-d}{j}^{-1}\right]^2\right)\binom{s}{j}V_{s,\omega}^j \\
&\quad + \frac{\binom{n-d}{s}}{\binom{n-1}{s-1}\binom{n}{s}}\left(\frac{\binom{n}{s}}{\binom{n-d}{s}}\sum_{a=1}^{d}\frac{\binom{d}{a}\binom{n-d}{s-a}}{\binom{n}{s}} + \left[1-\binom{n}{s}\binom{n-d}{s}^{-1}\right]^2\right)V_s^s\Bigg\} \\
&= \frac{d}{n-d}V_{s,\omega}^1 + \frac{n}{sd}\Bigg\{\sum_{j=2}^{d}\frac{\binom{s-1}{j-1}\binom{n-d}{j}}{\binom{n-1}{j-1}\binom{n}{j}}\left(\binom{n}{j}^2\binom{n-d}{j}^{-2} - \binom{n}{j}\binom{n-d}{j}^{-1}\right)\binom{s}{j}V_{s,\omega}^j \\
&\quad + \sum_{j=d+1}^{s-1}\frac{\binom{s-1}{j-1}\binom{n-d}{j}}{\binom{n-1}{j-1}\binom{n}{j}}\left(\frac{\binom{n}{j}}{\binom{n-d}{j}\binom{n}{d}}\sum_{a=1}^{d}\binom{j}{a}\binom{n-j}{d-a} + \left[1-\binom{n}{j}\binom{n-d}{j}^{-1}\right]^2\right)\binom{s}{j}V_{s,\omega}^j \\
&\quad + \frac{\binom{n-d}{s}}{\binom{n-1}{s-1}\binom{n}{s}}\left(\frac{\binom{n}{s}}{\binom{n-d}{s}\binom{n}{d}}\sum_{a=1}^{d}\binom{s}{a}\binom{n-s}{d-a} + \left[1-\binom{n}{s}\binom{n-d}{s}^{-1}\right]^2\right)V_s^s\Bigg\}
\end{aligned} \tag{A.11}$$

Next, we use the Chu-Vandermonde to simplify to the third and fourth summands. More

explicitly, observe the following.

$$\begin{aligned}\sum_{a=1}^{d}\binom{j}{a}\binom{n-j}{d-a} &= \binom{n}{d} - \binom{j}{0}\binom{n-j}{d} = \binom{n}{d} - \binom{n-j}{d}\\ \sum_{a=1}^{d}\binom{s}{a}\binom{n-s}{d-a} &= \binom{n}{d} - \binom{s}{0}\binom{n-s}{d} = \binom{n}{d} - \binom{n-s}{d}\end{aligned} \tag{A.12}$$

Plugging in gives us the following.

$$\begin{aligned}\mathbb{E}\left[(T_\ell)^2\right] = \frac{d}{n-d}V^1_{s,\omega} + \frac{n}{sd}\Bigg\{ &\sum_{j=2}^{d}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}}\left(\binom{n}{j}\binom{n-d}{j}^{-1} - 1\right)\binom{s}{j}V^j_{s,\omega}\\ &+ \sum_{j=d+1}^{s-1}\frac{\binom{s-1}{j-1}\binom{n-d}{j}}{\binom{n-1}{j-1}\binom{n}{j}}\left(\frac{\binom{n}{j}}{\binom{n-d}{j}}\left[1-\binom{n-j}{d}\binom{n}{d}^{-1}\right] + \left[1-\binom{n}{j}\binom{n-d}{j}^{-1}\right]^2\right)\binom{s}{j}V^j_{s,\omega}\\ &+ \frac{\binom{n-d}{s}}{\binom{n-1}{s-1}\binom{n}{s}}\left(\frac{\binom{n}{s}}{\binom{n-d}{s}}\left[1-\binom{n-s}{d}\binom{n}{d}^{-1}\right] + \left[1-\binom{n}{s}\binom{n-d}{s}^{-1}\right]^2\right)V^s_s\Bigg\}\end{aligned} \tag{A.13}$$

$$\begin{aligned}\mathbb{E}\left[(T_\ell)^2\right] = \frac{d}{n-d}V^1_{s,\omega} + \frac{n}{sd}\Bigg\{ &\sum_{j=2}^{d}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}}\left(\binom{n}{j}\binom{n-d}{j}^{-1} - 1\right)\binom{s}{j}V^j_{s,\omega}\\ &+ \sum_{j=d+1}^{s-1}\frac{\binom{s-1}{j-1}\binom{n-d}{j}}{\binom{n-1}{j-1}\binom{n}{j}}\left(\frac{\binom{n}{j}}{\binom{n-d}{j}} - 1 + \left[1-\binom{n}{j}\binom{n-d}{j}^{-1}\right]^2\right)\binom{s}{j}V^j_{s,\omega}\\ &+ \frac{\binom{n-d}{s}}{\binom{n-1}{s-1}\binom{n}{s}}\left(\frac{\binom{n}{s}}{\binom{n-d}{s}} - 1 + \left[1-\binom{n}{s}\binom{n-d}{s}^{-1}\right]^2\right)V^s_s\Bigg\}\end{aligned} \tag{A.14}$$

For a further simplification, make the following observation, where we use $a = \frac{\binom{n}{j}}{\binom{n-d}{j}}$.

$$a^{-1}\left(a - 1 + (1-a)^2\right) = \frac{a - 1 + 1 - 2a + a^2}{a} = \frac{a^2 - a}{a} = a - 1 \tag{A.15}$$

Plugging in leaves us with a considerably more simple equation.

$$\begin{aligned}\mathbb{E}\left[(T_\ell)^2\right] = \frac{d}{n-d}V^1_{s,\omega} + \frac{n}{sd}\Bigg\{ &\\ \sum_{j=2}^{s-1}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}}\left(\binom{n}{j}\binom{n-d}{j}^{-1} - 1\right)\binom{s}{j}V^j_{s,\omega} &+ \binom{n-1}{s-1}^{-1}\left(\binom{n}{s}\binom{n-d}{s}^{-1} - 1\right)V^s_s\Bigg\}\end{aligned} \tag{A.16}$$

As a next step, we want to show that the second case for the error term results in a functionally identical expression. Thus, we follow a very similar series of steps starting with the last

expression we derived.

$$\begin{aligned}\mathbb{E}\left[(T_\ell^\star)^2\right] &= \frac{d}{n-d}V_{s,\omega}^1 \\ &+ \frac{n}{sd}\left\{\sum_{j=2}^{s-1}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}\binom{n}{j}}\left(\binom{n}{j}-\binom{n-d}{j}+\left[1-\binom{n}{j}\binom{n-d}{j}^{-1}\right]^2\binom{n-d}{j}\right)\binom{s}{j}V_{s,\omega}^j\right. \\ &\left.+\binom{n-1}{s-1}^{-1}\binom{n}{s}^{-1}\left(\binom{n}{s}-\binom{n-d}{s}+\left[1-\binom{n}{s}\binom{n-d}{s}^{-1}\right]^2\binom{n-d}{s}\right)V_s^s\right\}\end{aligned} \tag{A.17}$$

Recognizing the similarity to a previous expression, we can observe the following.

$$1-a^{-1}+(1-a)^2a^{-1} = 1-a^{-1}+\frac{1-2a+a^2}{a} = 1-a^{-1}+a^{-1}-2+a = a-1 \tag{A.18}$$

Thus, we find the following.

$$\begin{aligned}\mathbb{E}\left[(T_\ell^\star)^2\right] &= \frac{d}{n-d}V_{s,\omega}^1+\frac{n}{sd}\left\{\right. \\ &\left.\sum_{j=2}^{s-1}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}}\left(\binom{n}{j}\binom{n-d}{j}^{-1}-1\right)\binom{s}{j}V_{s,\omega}^j+\binom{n-1}{s-1}^{-1}\left(\binom{n}{s}\binom{n-d}{s}^{-1}-1\right)V_s^s\right\}\end{aligned} \tag{A.19}$$

This shows that the two cases we created are essentially the same, allowing us to continue with the analysis without considering the distinction further. Next, we use the following.

$$\begin{aligned}\binom{n}{j}\binom{n-d}{j}^{-1} &= \binom{n}{j}\left(\binom{n}{j}\prod_{i=0}^{d-1}\frac{n-i-j}{n-i}\right)^{-1} = \prod_{i=0}^{d-1}\frac{n-i}{n-i-j} = \prod_{i=0}^{d-1}\left(1+\frac{j}{n-i-j}\right) \\ &\lesssim \left(1-\sum_{i=0}^{d-1}\frac{j}{n-i-j}\right)^{-1}\end{aligned} \tag{A.20}$$

Here, the last step follows from a Weierstrass inequality, found for example in [9]. Recall that $sd = o(n)$ and thus $\frac{jd}{n-j-d+1} \leq 1/2$ for n large enough and $s+d = o(n)$. This allows us to use a convenient upper bound as follows.

$$\begin{aligned}\binom{n}{j}\binom{n-d}{j}^{-1}-1 &\lesssim \frac{1-\left(1-\sum_{i=0}^{d-1}\frac{j}{n-i-j}\right)}{1-\sum_{i=0}^{d-1}\frac{j}{n-i-j}} = \frac{\sum_{i=0}^{d-1}\frac{j}{n-i-j}}{1-\sum_{i=0}^{d-1}\frac{j}{n-i-j}} \\ &\lesssim \frac{2jd}{n-j-d+1} \lesssim \frac{3jd}{n}\end{aligned} \tag{A.21}$$

Additionally, we continue by bounding this term of interest using a number of inequalities concerning the binomial coefficient. Specifically, we apply the following inequalities to transform binomial coefficients into more easily handled exponential expressions.

$$\max\left\{\left(\frac{n}{k}\right)^k, \frac{(n-k-1)^k}{k!}\right\} \leq \binom{n}{k} \leq \frac{n^k}{k!} \leq \left(\frac{en}{k}\right)^k \tag{A.22}$$

Plugging in these results gives us the following bound on our error terms.

$$\begin{aligned}
\mathbb{E}\left[(T_\ell)^2\right] &\lesssim \frac{d}{n-d}V_{s,\omega}^1 + \frac{n}{sd}\left\{\sum_{j=2}^{s-1}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}}\frac{3jd}{n}\binom{s}{j}V_{s,\omega}^j + \binom{n-1}{s-1}^{-1}\frac{3sd}{n}V_s^s\right\} \\
&\leq \frac{d}{n-d}V_{s,\omega}^1 + \frac{3n}{sd}\left\{\sum_{j=2}^{s-1}\left(\frac{e(s-1)}{n-1}\right)^{j-1}\frac{jd}{n}\binom{s}{j}V_{s,\omega}^j + \left(\frac{e(s-1)}{n-1}\right)^{s-1}\frac{sd}{n}V_s^s\right\} \\
&= \frac{d}{n-d}V_{s,\omega}^1 + \frac{3}{s}\left\{\sum_{j=2}^{s-1} j\left(\frac{e(s-1)}{n-1}\right)^{j-1}\binom{s}{j}V_{s,\omega}^j + s\left(\frac{e(s-1)}{n-1}\right)^{s-1}V_s^s\right\}
\end{aligned} \tag{A.23}$$

Proceeding this way allows us to continue our analysis by working towards a dominance argument for the Hájek projection terms. We begin by splitting the sum into two parts and applying a simple upper bound to the first. Then we employ a rather weak bound by replacing $V_{s,\omega}^j$ and V_s^s by ζ_s^s in the second part.

$$\begin{aligned}
\mathbb{E}\left[(T_\ell)^2\right] &\lesssim \frac{d}{n-d}V_{s,\omega}^1 + \frac{3}{s}\left\{\sum_{j=2}^{s-1} j\left(\frac{e(s-1)}{n-1}\right)^{j-1}\binom{s}{j}V_{s,\omega}^j + s\left(\frac{e(s-1)}{n-1}\right)^{s-1}V_s^s\right\} \\
&\leq \frac{d}{n-d}V_{s,\omega}^1 + \frac{3e(s-1)}{s(n-1)}\left\{\sum_{j=2}^{s-1}\binom{s}{j}V_{s,\omega}^j + V_s^s\right\} \\
&\quad + \frac{3}{s}\left\{\sum_{j=2}^{s-1}(j-1)\left(\frac{e(s-1)}{n-1}\right)^{j-1}\binom{s}{j}V_{s,\omega}^j + (s-1)\left(\frac{e(s-1)}{n-1}\right)^{s-1}V_s^s\right\} \\
&\leq \frac{d}{n-d}\zeta_{s,\omega}^1 + \frac{3e(s-1)}{s(n-1)}\left(\zeta_s^s - s\zeta_{s,\omega}^1\right) + \frac{3}{s}\zeta_s^s\left\{\sum_{j=2}^{s}(j-1)\left(\frac{e(s-1)}{n-1}\right)^{j-1}\right\} \\
&= \frac{d}{n-d}\zeta_{s,\omega}^1 + \frac{3e(s-1)}{s(n-1)}\left(\zeta_s^s - s\zeta_{s,\omega}^1\right) + \frac{3}{s}\zeta_s^s\sum_{j=1}^{s-1} j\left(\frac{e(s-1)}{n-1}\right)^{j}
\end{aligned} \tag{A.24}$$

Next, we extend the last sum to infinity and evaluate the term.

$$\begin{aligned}
\mathbb{E}\left[(T_\ell)^2\right] &\lesssim \frac{d}{n-d}\zeta_{s,\omega}^1 + \frac{3e(s-1)}{s(n-1)}\left(\zeta_s^s - s\zeta_{s,\omega}^1\right) + \frac{3}{s}\zeta_s^s\sum_{j=1}^{\infty} j\left(\frac{e(s-1)}{n-1}\right)^{j} \\
&= \frac{d}{n-d}\zeta_{s,\omega}^1 + \frac{3e(s-1)}{s(n-1)}\left(\zeta_s^s - s\zeta_{s,\omega}^1\right) + \frac{3e(s-1)(n-1)}{s\left(n-1-e(s-1)\right)^2}\zeta_s^s \\
&= \left[\frac{d}{n-d} + \frac{3e(s-1)(n-1)}{\left(n-1-e(s-1)\right)^2}\right]\zeta_{s,\omega}^1 \\
&\quad + \frac{3e(s-1)}{s}\left[\frac{1}{n-1} + \frac{n-1}{\left(n-1-e(s-1)\right)^2}\right]\left(\zeta_s^s - s\zeta_{s,\omega}^1\right)
\end{aligned} \tag{A.25}$$

We continue by investigating the ratio of $\mathbb{E}\left[(T_\ell)^2\right]$ to $\zeta_{s,\omega}^1$ to show that the error term becomes

asymptotically negligible.

$$\begin{aligned}\frac{\mathbb{E}\left[T_\ell^2\right]}{\zeta_{s,\omega}^1} \leq & \underbrace{\frac{d}{n-d}}_{\sim d/n \longrightarrow 0} + \underbrace{\frac{3e(s-1)(n-1)}{(n-1-e(s-1))^2}}_{\sim sn/(n-s)^2 \longrightarrow 0} \\ & + \underbrace{\frac{3e(s-1)}{s}}_{\longrightarrow 3e} \left[\underbrace{\frac{1}{n-1}}_{\sim 1/n} + \underbrace{\frac{n-1}{(n-1-e(s-1))^2}}_{\sim n/(n-s)^2} \right] \frac{\zeta_s^s - s\zeta_{s,\omega}^1}{\zeta_{s,\omega}^1} \\ \sim & \; 3e\left(\frac{1}{n} + \frac{n}{(n-s)^2}\right) \frac{\zeta_s^s - s\zeta_{s,\omega}^1}{\zeta_{s,\omega}^1}\end{aligned} \tag{A.26}$$

Now, we will use the assumption that $\frac{s}{n}\left(\frac{\zeta_s^s}{s\zeta_{s,\omega}^1} - 1\right) \longrightarrow 0$.

$$\frac{\mathbb{E}\left[T_\ell^2\right]}{\zeta_{s,\omega}^1} \sim \underbrace{3e\frac{n}{s}\left(\frac{1}{n} + \frac{n}{(n-s)^2}\right)}_{O(1)} \underbrace{\frac{s}{n}\frac{\zeta_s^s - s\zeta_{s,\omega}^1}{\zeta_{s,\omega}^1}}_{\longrightarrow 0} \longrightarrow 0 \tag{A.27}$$

Therefore, we can conclude that $\frac{1}{\sqrt{d}}\sum_{i\in\ell} h_{s,\omega}^{(1)}(Z_i)$ dominates T_ℓ in the expression of interest. To use Lemma A.1, we need to furthermore show that we obtain convergence in probability for the terms corresponding to the X_i^2 in Lemma A.1. This is not immediate in the case of the delete-d jackknife, but can be shown as follows.

$$\begin{aligned}&\binom{n}{d}^{-1} \sum_{\ell\in L_{n,d}} \left(\frac{1}{\sqrt{d}}\sum_{i\in\ell} h_{s,\omega}^{(1)}(Z_i)\right)^2 = d^{-1}\binom{n}{d}^{-1} \sum_{\ell\in L_{n,d}} \sum_{i\in\ell}\sum_{j\in\ell} h_{s,\omega}^{(1)}(Z_i)h_{s,\omega}^{(1)}(Z_j) \\ &= d^{-1}\binom{n}{d}^{-1} \sum_{\ell\in L_{n,d}} \sum_{i\in\ell} \left(h_{s,\omega}^{(1)}(Z_i)\right)^2 = d^{-1}\binom{n}{d}^{-1} \sum_{i=1}^n \sum_{\ell\in L_{n,d}} \mathbb{1}\left(i\in\ell\right)\left(h_{s,\omega}^{(1)}(Z_i)\right)^2 \\ &= d^{-1}\binom{n}{d}^{-1} \sum_{i=1}^n \binom{n-1}{d-1}\left(h_{s,\omega}^{(1)}(Z_i)\right)^2 \\ &= \frac{1}{n}\sum_{i=1}^n \left(h_{s,\omega}^{(1)}(Z_i)\right)^2 \longrightarrow_p \mathbb{E}\left[\left(h_{s,\omega}^{(1)}(Z_i)\right)^2\right] = \zeta_{1,\omega}^s\end{aligned} \tag{A.28}$$

Thus, using Lemma A.1 we can thus conclude the following.

$$\frac{n\widehat{\sigma}_{JKD}^2\left(D_{[n]};d\right)}{s^2\zeta_{s,\omega}^1} \longrightarrow_p \frac{n\binom{n}{d}^{-1}\sum_{\ell\in L_{n,d}}\left(\frac{1}{\sqrt{d}}\sum_{i\in\ell}\left(h_{s,\omega}^{(1)}(Z_i)\right)\right)^2}{s^2\zeta_{s,\omega}^1} \longrightarrow 1 \tag{A.29}$$

The desired rate-consistency then immediately follows from an application of Lemma 3.1. □

Proof of Theorem 3.4.

To extend our result to potentially incomplete generalized U-statistics, we follow an analogous approach to [12] Theorem 7. We start by defining a modified kernel that incorporates the randomness due to Bernoulli subsampling as a kernel feature.

$$\hbar_s\left(D_{[s]};\rho,\omega\right) = \frac{\rho}{p}h_s\left(D_{[s]};\omega\right) \tag{A.30}$$

We define an auxiliary complete generalized U-statistic based on this modified kernel.

$$\overline{U}_{n,s,\rho,\omega} = \binom{n}{s}^{-1} \sum_{\ell \in L_{n,s}} \hbar_s \left(D_\ell ; \rho_\ell, \omega \right) \tag{A.31}$$

We then proceed by comparing the generalized Hoeffding decomposition of $\overline{U}_{n,s,\rho,\omega}$ to its original unmodified counterpart.

Remark 4. Due to the structure of this modified estimator, it can also be conceived as a Horvitz-Thompson normalized version of the original incomplete generalized U-statistic. As part of the following proof, we also show consistency of the Jackknife variance estimator for this Horvitz-Thompson normalized version; a result that could be of interest in itself.

Due to the construction of the generalized Hoeffding decomposition, we can make an immediate observation about terms that are shared between the two. Namely, for $1 \leq i \leq s-1$, we have the following.

$$\hbar_s^{(i)} \left(D_{[i]} \right) = h_s^{(i)} \left(D_{[i]} \right) \tag{A.32}$$

This naturally extends to the terms H_s^i, $V_{s,\omega}^i$, $\zeta_{s,\omega}^i$ and their modified counterparts. Thus, when considering the generalized Hoeffding decomposition, the difference between the two only appears in their respective final terms of each type. We thus introduce the notation $\overline{V}_s^s$ and $\overline{\zeta}_s^s$ for the modified equivalents of V_s^s and ζ_s^s to indicate this difference. As a first consequence of this observation, we find the following variance decomposition for the modified kernel.

$$\overline{\zeta}_s^s = \operatorname{Var}\left(\hbar_s \left(D_{[s]} ; \rho, \omega \right) \right) = \sum_{j=1}^{s-1} \binom{s}{j} V_{s,\omega}^j + \overline{V}_s^s \tag{A.33}$$

Clearly, to understand the impact of the Bernoulli sampling scheme a closer analysis of $\overline{V}_s^s$ is necessary. Thus, observe the following.

$$\overline{\zeta}_s^s = \operatorname{Var}\left(\hbar_s \left(D_{[n]} ; \rho, \omega \right) \right) = \operatorname{Var}\left(\frac{\rho}{p} h_s \left(D_{[s]} ; \omega \right) \right) = \frac{1-p}{p} \zeta_s^s + \zeta_s^s \tag{A.34}$$

Thus, we can deduce that $\overline{V}_s^s = V_s^s + \frac{1-p}{p} \zeta_s^s$. In combination with an earlier idea, the separation of the delete-d Jackknife variance estimator into a Hájek term and a residual term, we can now adjust for the changes in the residual term due to the Bernoulli sampling scheme.

$$\begin{aligned}
\widehat{\sigma}_{JKD}^2 \left(D_{[n]} ; d \right) &= \frac{n-d}{d} \binom{n}{d}^{-1} \left(\frac{s}{n} \right)^2 \sum_{\ell \in L_{n,d}} \Bigg[\sum_{i \in \ell} h_{s,\omega}^{(1)}(Z_i) - \frac{d}{n-d} \sum_{i \in [n] \setminus \ell} h_{s,\omega}^{(1)}(Z_i) \\
&\quad + \frac{n}{s} \sum_{j=2}^{s-1} \binom{s}{j} \Bigg(\binom{n}{j}^{-1} \sum_{\iota \in L_{n,j}} h_{s,\omega}^{(j)}(D_\iota) - \binom{n-d}{j}^{-1} \sum_{\iota \in L_j([n] \setminus \ell)} h_{s,\omega}^{(j)}(D_\iota) \Bigg) \\
&\quad + \binom{n}{s}^{-1} \sum_{\iota \in L_{n,s}} h_s^{(s)}(D_\iota) - \binom{n-d}{s}^{-1} \sum_{\iota \in L_s([n] \setminus \ell)} \hbar_s^{(s)}(D_\iota) \Bigg]^2 \\
&=: \frac{s^2}{(n-d)n^2} \binom{n}{d}^{-1} \sum_{\ell \in L_{n,d}} \left[\frac{1}{\sqrt{d}} \sum_{i \in \ell} h_{s,\omega}^{(1)}(Z_i) + \overline{T}_\ell \right]^2
\end{aligned} \tag{A.35}$$

Using this observation, we can recall an earlier result, namely Equation A.23, and adjust for the changes in the final variance term.

$$\begin{aligned}
\mathbb{E}\left[\left(\overline{T}_\ell\right)^2\right] &\lesssim \frac{d}{n-d}V^1_{s,\omega} \\
&\quad + \frac{n}{sd}\left\{\sum_{j=2}^{s-1}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}}\frac{3jd}{n}\binom{s}{j}V^j_{s,\omega} + \binom{n-1}{s-1}^{-1}\frac{3sd}{n}\left(V^s_s + \frac{1-p}{p}\zeta^s_s\right)\right\} \\
&= \frac{d}{n-d}V^1_{s,\omega} \\
&\quad + \frac{n}{sd}\left\{\sum_{j=2}^{s-1}\frac{\binom{s-1}{j-1}}{\binom{n-1}{j-1}}\frac{3jd}{n}\binom{s}{j}V^j_{s,\omega} + \binom{n-1}{s-1}^{-1}\frac{3sd}{n}V^s_s\right\} + 3\binom{n-1}{s-1}^{-1}\frac{1-p}{p}\zeta^s_s
\end{aligned} \tag{A.36}$$

In essence, we end up with an adjustment term in addition to the terms we encountered in the complete case. We need to understand how this term behaves asymptotically relative to $\zeta^1_{s,\omega}$ as we rely on the dominance of the latter.

$$\begin{aligned}
3\binom{n-1}{s-1}^{-1}\frac{1-p}{p}\frac{\zeta^s_s}{\zeta^1_{s,\omega}} &= 3\binom{n-1}{s-1}^{-1}\left(\frac{\binom{n}{s}}{N} - 1\right)\frac{\zeta^s_s}{\zeta^1_{s,\omega}} \\
&= 3\left(\frac{n}{Ns} - \binom{n-1}{s-1}^{-1}\right)\frac{\zeta^s_s}{\zeta^1_{s,\omega}} \sim 3\zeta^s_s\frac{n}{Ns\zeta^1_{s,\omega}}
\end{aligned} \tag{A.37}$$

By the boundedness of ζ^s_s and the asymptotically-sufficient sampling condition (Assumption 2), this last term goes to 0. Thus, the delete-d jackknife is ratio-consistent for the variance of $\overline{U}_{n,s,\rho,\omega}$ in the sense of Theorem 3.3. It remains to be shown that the normalization by $\widehat{N}$ instead of N preserves this property to obtain ratio-consistency for the variance of the actual incomplete generalized U-statistic. To illustrate this desired property, it is beneficial to make explicit use of θ_s to address the implicit centering that obscures the potential discrepancy.

$$\begin{aligned}
U_{n,s,N,\omega} - \overline{U}_{n,s,\rho,\omega} &= \widehat{N}^{-1}\sum_{\ell\in L_{n,s}}\rho_\ell h_s\left(D_\ell;\omega\right) - \binom{n}{s}^{-1}\sum_{\ell\in L_{n,s}}\frac{\rho_\ell}{p}h_s\left(D_\ell;\omega\right) \\
&= \widehat{N}^{-1}\sum_{\ell\in L_{n,s}}\rho_\ell\left(h_s\left(D_\ell;\omega\right) - \theta_s\right) - N^{-1}\sum_{\ell\in L_{n,s}}\rho_\ell\left(h_s\left(D_\ell;\omega\right) - \theta_s\right) \\
&= \widehat{N}^{-1}\sum_{\ell\in L_{n,s}}\left(1 - \frac{\widehat{N}}{N}\right)\rho_\ell\left(h_s\left(D_\ell;\omega\right) - \theta_s\right)
\end{aligned} \tag{A.38}$$

We want to use Lemma A.1 to control this deviation. Thus observe the following.

$$
\begin{aligned}
&\mathbb{E}\left[\left(\left(1-\frac{\widehat{N}}{N}\right)\rho_\ell\left(h_s\left(D_\ell;\omega\right)-\theta_s\right)\right)^2\right] \\
&\quad=\zeta_s^s\mathbb{E}\left[\left(1-\frac{\widehat{N}}{N}\right)^2\rho_{[s]}\right]\mathbb{E}\left[\left(h_s\left(D_\ell;\omega\right)-\theta_s\right)^2\right]=\frac{\zeta_s^s}{N^2}\mathbb{E}\left[\left(N-\widehat{N}\right)^2\rho_{[s]}\right] \\
&\quad=\frac{\zeta_s^s}{N^2}\mathbb{E}\left[\left((N-1)-\left(\widehat{N}-\rho_{[s]}\right)+\left(1-\rho_{[s]}\right)\right)^2\rho_{[s]}\right] \\
&\quad=\frac{\zeta_s^s}{N^2}\Bigg\{\mathbb{E}\left[\left((N-1)-\left(\widehat{N}-\rho_{[s]}\right)\right)^2\rho_{[s]}\right] \\
&\qquad-2\mathbb{E}\left[\left((N-1)-\left(\widehat{N}-\rho_{[s]}\right)\right)\left(1-\rho_{[s]}\right)\rho_{[s]}\right]+\mathbb{E}\left[\left(1-\rho_{[s]}\right)^2\rho_{[s]}\right]\Bigg\} \\
&\quad=\zeta_s^s\frac{p}{N^2}\mathbb{E}\left[\left(\left(\widehat{N}-\rho_{[s]}\right)-(N-1)\right)^2\right]=\zeta_s^s\frac{p}{N^2}(N-1)p(1-p) \\
&\quad=\zeta_s^s\frac{N-1}{N\binom{n}{s}}\frac{N}{\binom{n}{s}}\frac{\binom{n}{s}-N}{\binom{n}{s}}\lesssim\zeta_s^s\binom{n}{s}^{-1}\longrightarrow 0
\end{aligned}
\tag{A.39}
$$

The convergence to zero follows from the fact that ζ_s^s is bounded. Thus, the potential discrepancies are dominated by their corresponding terms in $\overline{U}_{n,s,\rho,\omega}$ in the sense of Lemma A.1, and we obtain the desired result by an application of said Lemma. □

Appendix B: Proofs for Application to TDNN

B.1. Notation regarding the (T)DNN estimators

As the projections and corresponding variance terms differ for the DNN and TDNN estimators, it is useful to explicitly define the necessary terms here to avoid any confusion going forward. We start by once again introducing the relevant projection kernels for $c = 2, \ldots, s$ and $c = 2, \ldots, s_2$, respectively.

$$\begin{aligned} \psi_s^c(x; d_{[c]}) &= \mathbb{E}\left[h_s\left(x; D_{[s]}\right) \middle| D_{[c]} = d_{[c]}\right] \quad \text{and} \\ \psi_{\mathfrak{S}}^c(x; d_{[c]}) &= \mathbb{E}\left[h_{\mathfrak{S}}\left(x; D_{[s_2]}\right) \middle| D_{[c]} = d_{[c]}\right] \end{aligned} \tag{B.1}$$

$$h_s^{(1)}(x; d_1) = \psi_s^1(x; d_1) - \mu(x) \quad \text{and} \quad h_{\mathfrak{S}}^{(1)}(x; d_1) = \psi_{\mathfrak{S}}^1(x; d_1) - \mu(x) \tag{B.2}$$

Recursively, we define for $c = 2, \ldots, s$ and $c = 2, \ldots, s_2$, respectively, in the following way.

$$\begin{aligned} h_s^{(c)}\left(x; d_{[c]}\right) &= \psi_s^c(x; d_{[c]}) - \sum_{j=1}^{c-1}\left(\sum_{\ell \in L_{c,j}} h_s^{(j)}(x; d_\ell)\right) - \mu(x) \\ h_{\mathfrak{S}}^{(c)}\left(x; d_{[c]}\right) &= \psi_{\mathfrak{S}}^c(x; d_{[c]}) - \sum_{j=1}^{c-1}\left(\sum_{\ell \in L_{c,j}} h_{\mathfrak{S}}^{(j)}(x; d_\ell)\right) - \mu(x) \end{aligned} \tag{B.3}$$

The main purpose of this additional notation is to derive the Hoeffding decompositions of the DNN and TDNN estimator that will be essential in the proofs of the main theorems. Considering the projections for $c = 1, \ldots, s$ and $c = 1, \ldots, s_2$, respectively, in accordance with Definition 2.2, we use the following notation.

$$\begin{aligned} \tilde{\mu}_s\left(x; D_{[n]}\right) &= \mu(x) + \sum_{j=1}^{s}\binom{s}{j}\binom{n}{j}^{-1}\sum_{\ell \in L_{n,j}} h_s^{(j)}(x; D_\ell) \\ \widehat{\mu}_{\mathfrak{S}}\left(x; D_{[n]}\right) &= \mu(x) + \sum_{j=1}^{s_2}\binom{s_2}{j}\binom{n}{j}^{-1}\sum_{\ell \in L_{n,j}} h_{\mathfrak{S}}^{(j)}(x; D_\ell) \end{aligned} \tag{B.4}$$

At this point, we will also introduce a number of covariance terms that will be crucial for the analysis of the DNN and TDNN kernels. For the DNN estimator and any $1 \le c \le s$, define

$$\xi_s^c(x) = \mathrm{Var}\left(\psi_s^c(x; D_{[c]})\right) \tag{B.5}$$

$$\Omega_s(x) = \mathbb{E}\left[h_s^2\left(x; D_{[s]}\right)\right] \tag{B.6}$$

$$\Omega_s^c(x) = \mathbb{E}\left[h_s\left(x; D_{[s]}\right) \cdot h_s\left(x; D'_{[s]}\right)\right] \tag{B.7}$$

where as before $D_{[s]} = \{Z_1, \ldots, Z_s\}$ be a vector of i.i.d. random variables drawn from P and $D'_{[s]} = \{Z_1, \ldots, Z_c, Z'_{c+1}, \ldots, Z'_s\}$ where $Z'_{c+1}, \ldots, Z'_s$ are i.i.d. draws from P that are independent of $D_{[s]}$. Similarly, for the TDNN estimator, we define the following variance-covariance terms for given $0 < s_1 < s_2 < n$ and $c \le s_1$.

$$\zeta_{\mathfrak{S}}^{s_2}(x) = \mathrm{Var}\left(h_{\mathfrak{S}}\left(x; D_{[s_2]}\right)\right) \tag{B.8}$$

$$\Upsilon_{\mathfrak{S}}(x) = \mathbb{E}\left[h_{s_1}\left(x; D_{[s_1]}\right) \cdot h_{s_2}\left(x; D_{[s_2]}\right)\right] \tag{B.9}$$

$$\Upsilon_{\mathfrak{S}}^c(x) = \mathbb{E}\left[h_{s_1}\left(x; D'_{[s_1]}\right) \cdot h_{s_2}\left(x; D_{[s_2]}\right)\right] \tag{B.10}$$

Here, $D_{[s_2]} = \{Z_1, \ldots, Z_{s_2}\}$ is a vector of i.i.d. random variables drawn from P for $s_2 > s_1$. Furthermore, we let $D'_{[s_1]} = \{Z_1, \ldots, Z_c, Z'_{c+1}, \ldots, Z'_{s_1}\}$ where $Z'_{c+1}, \ldots, Z'_{s_1}$ are i.i.d. draws from P that are independent of $D_{[s_2]}$.

B.2. Lemmas concerning the DNN kernel

Throughout the proofs in this section, we are concerned with the same setup. For the sake of brevity, we introduce this setup here to avoid unnecessary repetition in the following lemmas. Consider a sample size n, a subsampling scale s growing with n, and c such that $0 < c \leq s \leq n$. Let $D_{[s]} = \{Z_1, Z_2, \ldots, Z_c, Z_{c+1}, \ldots Z_s\}$ be an i.i.d. data set drawn from P as described in Setup 3. Let $D'_{[s]} = \left\{Z_1, Z_2, \ldots, Z_c, Z'_{c+1}, \ldots Z'_s\right\}$ be a second data set that shares the first c observations with $D_{[s]}$. The remaining $s-c$ observations of $D'_{[s]}$, i.e. $\left\{Z'_{c+1}, \ldots Z'_s\right\}$, are i.i.d. draws from P that are independent of $D_{[s]}$. All expectations are with respect to all random elements unless a conditioning bar is displayed; we write $\mathbb{E}\left[\cdot|X\right]$ for conditional expectation given the sigma-field generated by the displayed variables.

Lemma B.1 ([6] - Lemma 12)**.**
The indicator functions $\kappa\left(x; Z_i, D_{[s]}\right)$ satisfy the following properties.

1. *For any $i \neq j$, we have $\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z_j, D_{[s]}\right) = 0$ with probability one;*
2. $\sum_{i=1}^{s} \kappa\left(x; Z_i, D_{[s]}\right) = 1$;
3. $\forall i \in [s]: \quad \mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\right] = s^{-1}$
4. $\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\middle| D_1 = Z_1\right] = \left\{1 - \varphi\left(B\left(x, \|X_1 - x\|\right)\right)\right\}^{s-1}$

Lemma B.2 ([6] - Lemma 13)**.**
For any L^1 function f that is continuous at x, it holds that

$$\lim_{s \longrightarrow \infty} \mathbb{E}\left[f\left(X_1\right) s \mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\middle| X_1\right]\right] = f(x). \tag{B.11}$$

As an additional tool, we will make use of the following analogous results concerning products of two kernel functions with nonzero expectation. These results will then be used to construct an analog of Lemma B.2 for the corresponding cases. This will serve very similar purposes in the analysis of (conditional) covariance terms as the previous results serve for (conditional) expectations.

Lemma B.3.
The following three statements hold.

$$\forall i \in [c]: \quad \mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z_i, D'_{[s]}\right)\right] = (2s-c)^{-1} = \omega(e^{-s}) \tag{B.12}$$

$$\begin{aligned}\forall i \in [c]\ \forall j \in \{c+1, \ldots, s\}: \quad & \mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z'_j, D'_{[s]}\right)\right] \\ &= \frac{1}{(2s-c)(2s-c-1)} \sum_{i=0}^{s-c-1} \binom{s-c-1}{i}\binom{2s-c-2}{i}^{-1} = \omega(e^{-s})\end{aligned} \tag{B.13}$$

$$\begin{aligned}\forall i, j \in \{c+1, \ldots, s\}: \quad & \mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z'_j, D'_{[s]}\right)\right] \\ &= \frac{2}{(2s-c)(2s-c-1)} \sum_{i=0}^{s-c-1} \binom{s-c-1}{i}\binom{2s-c-1}{s-1+i}^{-1} = \omega(e^{-s})\end{aligned} \tag{B.14}$$

Proof of Lemma B.3.
Without loss of generality, we will consider the cases of $i = 1$ and $j = c + 1$ for the first two equations.

$$\begin{aligned}
&\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z_1, D'_{[s]}\right)\right] \\
&\quad = \mathbb{E}\left[\kappa\left(x; Z_1, D_{[c]}\right)\kappa\left(x; Z_1, D_{(c+1):s}\right)\kappa\left(x; Z_1, D'_{(c+1):s}\right)\right] \\
&\quad = \mathbb{E}\left[\kappa\left(x; Z_1, D_{[2s-c]}\right)\right] = (2s-c)^{-1}
\end{aligned} \tag{B.15}$$

Considering the second case, we find the following.

$$\begin{aligned}
&\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right] \\
&= \frac{1}{(2s-c)!}\sum_{i=0}^{s-c-1}\binom{s-c-1}{i} i!\,((s-1)+(s-c-1-i))! \\
&= \frac{1}{(2s-c)!}\sum_{i=0}^{s-c-1}\binom{s-c-1}{i} i!\,(2s-c-2-i)! \\
&= \frac{(2s-c-2)!}{(2s-c)!}\sum_{i=0}^{s-c-1}\binom{s-c-1}{i}\binom{2s-c-2}{i}^{-1}
\end{aligned} \tag{B.16}$$

While unintuitive at first, the terms in this expression have intuitive meaning when we consider this as a combinatorial problem. Consider lining up the observations in order of their distance to the point of interest and counting the cases for which the expression in the expectation is equal to one. First, there are $(2s-c)!$ possible orderings of the observations with probability one, leading to the denominator. Next, notice that only those orderings where $\|X'_{c+1} - x\| \leq \|X_1 - x\|$ and $\|X_1 - x\| \leq \|X_i - x\|$ for any $i = 2, \ldots, c$ can possibly lead to a non-zero realization of the kernel term. Furthermore, out of the $(s-c-1)$ observations in $D'_{(c+2):s}$, it is possible for $i = 0, \ldots, s-c-1$ observations to lie at a distance to the point of interest that is smaller than $\|X_1 - x\|$ but larger than $\|X'_{c+1} - x\|$ in any permutation. The sum adjusts for those possible configurations. Next, we can make the following observation concerning the expression we just derived.

$$\begin{aligned}
&\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right] \\
&\geq \frac{1}{(2s-c)(2s-c-1)}\sum_{i=0}^{s-c-1}\frac{(s-c-i)^i}{i!}\frac{i!}{(2s-c-2)^i} \\
&= \frac{1}{(2s-c)(2s-c-1)}\left(1 + \sum_{i=0}^{s-c-1}\left(\frac{s-c-i}{2s-c-2}\right)^i\right) \geq \frac{1}{(2s-c)^2}
\end{aligned} \tag{B.17}$$

We can now observe the following using the small Omega Bachmann-Landau notation.

$$\lim_{s\longrightarrow\infty}\frac{(2s-c)^{-2}}{e^{-s}} = \infty \quad \Longrightarrow \quad \mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right] = \omega(e^{-x}) \tag{B.18}$$

Considering the third case, without loss of generality, we consider the case of $i = j = c+1$.

We find the following.

$$\begin{aligned}
&\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right] \\
&= \mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[c]}\right)\kappa\left(x; Z'_{c+1}, D_{[c]}\right)\kappa\left(x; Z_{c+1}, D_{(c+1):s}\right)\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\right] \\
&= \frac{2}{(2s-c)!}\sum_{i=0}^{s-c-1}\binom{s-c-1}{i}(s-1+i)!(s-c-1-i)! \\
&= \frac{2(2s-c-2)!}{(2s-c)!}\sum_{i=0}^{s-c-1}\binom{s-c-1}{i}\binom{2s-c-2}{s-1+i}^{-1} \\
&= \frac{2}{(2s-c)(2s-c-1)}\sum_{i=0}^{s-c-1}\binom{s-c-1}{i}\binom{2s-c-2}{s-1+i}^{-1}
\end{aligned} \tag{B.19}$$

The third case follows from a similar combinatorial logic as the second. We consider without loss of generality the case that $\|X'_{c+1} - x\| \leq \|X_{c+1} - x\|$ and adjust for this fact by multiplying the whole expression by two. Notice now that any number $i = 0, \ldots, s-c-1$ of observations in $D'_{(c+2):s}$ can be farther away from x than X_{c+1} or at a distance that is between $\|X'_{c+1} - x\|$ and $\|X_{c+1} - x\|$. The summation adjusts for all possible permutations that satisfy this criterion. Furthermore, we can make the following observation.

$$\begin{aligned}
&\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right] \geq \frac{1}{(2s-c)^2}\sum_{i=0}^{s-c-1}\binom{s-c-1}{i}\binom{2s-c-2}{s-1+i}^{-1} \\
&= \frac{1}{(2s-c)^2}\sum_{i=0}^{s-c-1}\left(\frac{(s-c-1)!}{i!(s-c-1-i)!}\frac{(s-1+i)!(s-c-1-i)!}{(2s-c-2)!}\right) \\
&= \frac{1}{(2s-c)^2}\frac{(s-c-1)!}{(2s-c-2)!}\sum_{i=0}^{s-c-1}\frac{(s-1+i)!}{i!} \\
&= \frac{1}{(2s-c)^2}\frac{(s-c-1)!}{(2s-c-2)!}\left(\frac{(2s-c-2)!}{(s-c-1)!} + \sum_{i=0}^{s-c-2}\frac{(s-1+i)!}{i!}\right) \\
&\quad \geq \frac{1}{(2s-c)^2}
\end{aligned} \tag{B.20}$$

We can now observe the following using the small Omega Bachmann-Landau notation.

$$\lim_{s\longrightarrow\infty}\frac{(2s-c)^{-2}}{e^{-s}} = \infty \quad \Longrightarrow \quad \mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right] = \omega(e^{-x}) \tag{B.21}$$

□

Lemma B.4.
The following statements hold.

$$\forall i \in [c]: \quad \mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z_i, D'_{[s]}\right)\middle| X_i\right] = \left\{1-\varphi\left(B\left(x, \|X_i - x\|\right)\right)\right\}^{2s-c-1} \tag{B.22}$$

$$\begin{aligned}\forall i \in [c]\ \forall j \in \{c+1,\ldots,s\}: \quad & \mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z'_j, D'_{[s]}\right)\middle| X_i, X'_j\right] \\ & = \mathbb{1}\left(\|X'_j - x\| \le \|X_i - x\|\right)\left\{1-\varphi\left(B\left(x, \|X_i - x\|\right)\right)\right\}^{s-1}\left\{1-\varphi\left(B\left(x, \left\|X'_j - x\right\|\right)\right)\right\}^{s-c-1}\end{aligned} \tag{B.23}$$

$$\begin{aligned}\forall i, j \in \{c+1,\ldots,s\}: \quad & \mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z'_j, D'_{[s]}\right)\middle| X_i, X'_j\right] \\ & = \left\{1-\varphi\left(B\left(x, \min\left(\|X_i - x\|, \left\|X'_j - x\right\|\right)\right)\right)\right\}^{s-c-1}\left\{1-\varphi\left(B\left(x, \max\left(\|X_i - x\|, \left\|X'_j - x\right\|\right)\right)\right)\right\}^{s-1}\end{aligned} \tag{B.24}$$

Proof of Lemma B.4.
Without loss of generality, we will consider the cases of $i = 1$ for the first equation.

$$\begin{aligned}& \mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z_1, D'_{[s]}\right)\middle| X_1\right] \\ & = \mathbb{E}\left[\kappa\left(x; Z_1, D_{[c]}\right)\kappa\left(x; Z_1, D_{(c+1):s}\right)\kappa\left(x; Z_1, D'_{(c+1):s}\right)\middle| X_1\right] \\ & = \mathbb{E}\left[\kappa\left(x; Z_1, D_{[c]}\right)\middle| X_1\right]\mathbb{E}\left[\kappa\left(x; Z_1, D_{(c+1):s}\right)\middle| X_1\right]\mathbb{E}\left[\kappa\left(x; Z_1, D'_{(c+1):s}\right)\middle| X_1\right] \\ & = \left\{1-\varphi\left(B\left(x, \|X_i - x\|\right)\right)\right\}^{c-1}\left\{1-\varphi\left(B\left(x, \|X_i - x\|\right)\right)\right\}^{s-c}\left\{1-\varphi\left(B\left(x, \|X_i - x\|\right)\right)\right\}^{s-c} \\ & = \left\{1-\varphi\left(B\left(x, \|X_i - x\|\right)\right)\right\}^{2s-c-1}\end{aligned} \tag{B.25}$$

Without loss of generality, we will consider the cases of $i = 1$ and $j = c + 1$ for the second equation.

$$
\begin{aligned}
&\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_1, X'_{c+1}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\kappa\left(x; Z_1, D_{[c]}\right)\kappa\left(x; Z_1, D_{(c+1):s}\right)\kappa\left(x; Z'_{c+1}, D_{[c]}\right)\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\middle| X_{[c]}, X'_{c+1}\right]\middle| X_1, X'_{c+1}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\kappa\left(x; Z_1, D_{(c+1):s}\right)\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\middle| X_{[c]}, X'_{c+1}\right]\kappa\left(x; Z_1, D_{[c]}\right)\kappa\left(x; Z'_{c+1}, D_{[c]}\right)\middle| X_1, X'_{c+1}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\kappa\left(x; Z_1, D_{(c+1):s}\right)\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\middle| X_1, X'_{c+1}\right]\kappa\left(x; Z_1, D_{[c]}\right)\kappa\left(x; Z'_{c+1}, D_{[c]}\right)\middle| X_1, X'_{c+1}\right] \\
&= \mathbb{E}\left[\kappa\left(x; Z_1, D_{(c+1):s}\right)\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\middle| X_1, X'_{c+1}\right]\mathbb{E}\left[\kappa\left(x; Z_1, D_{[c]}\right)\kappa\left(x; Z'_{c+1}, D_{[c]}\right)\middle| X_1, X'_{c+1}\right] \\
&= \mathbb{E}\left[\kappa\left(x; Z_1, D_{(c+1):s}\right)\middle| X_1\right]\mathbb{E}\left[\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\middle| X'_{c+1}\right]\mathbb{1}\left(\left\|X'_{c+1} - x\right\| \le \left\|X_1 - x\right\|\right)\mathbb{E}\left[\kappa\left(x; Z_1, D_{[c]}\right)\middle| X_1\right] \\
&= \mathbb{1}\left(\left\|X'_{c+1} - x\right\| \le \left\|X_1 - x\right\|\right)\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\middle| X_1\right]\mathbb{E}\left[\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\middle| X'_{c+1}\right] \\
&= \mathbb{1}\left(\left\|X'_{c+1} - x\right\| \le \left\|X_1 - x\right\|\right)\left\{1 - \varphi\left(B\left(x, \left\|X_1 - x\right\|\right)\right)\right\}^{s-1}\left\{1 - \varphi\left(B\left(x, \left\|X'_{c+1} - x\right\|\right)\right)\right\}^{s-c-1}
\end{aligned}
\tag{B.26}
$$

For the third case, without loss of generality, we consider the case of $i = j = c + 1$.

$$
\begin{aligned}
&\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_{c+1}, X'_{c+1}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_{[c]}, X_{c+1}, X'_{c+1}\right]\middle| X_{c+1}, X'_{c+1}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[c+1]}\right)\kappa\left(x; Z'_{c+1}, D'_{[c+1]}\right)\kappa\left(x; Z_{c+1}, D_{(c+1):s}\right)\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\middle| X_{[c]}, X_{c+1}, X'_{c+1}\right]\middle| X_{c+1}, X'_{c+1}\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[c+1]}\right)\kappa\left(x; Z'_{c+1}, D'_{[c+1]}\right)\middle| X_{[c]}, X_{c+1}, X'_{c+1}\right]\kappa\left(x; Z_{c+1}, D_{(c+1):s}\right)\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\middle| X_{c+1}, X'_{c+1}\right] \\
&= \mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[c+1]}\right)\kappa\left(x; Z'_{c+1}, D'_{[c+1]}\right)\middle| X_{c+1}, X'_{c+1}\right]\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{(c+1):s}\right)\middle| X_{c+1}\right]\mathbb{E}\left[\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\middle| X'_{c+1}\right]
\end{aligned}
\tag{B.27}
$$

Without loss of generality, consider the case that $\|X_{c+1} - x\| \leq \|X'_{c+1} - x\|$.

$$\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[c+1]}\right)\kappa\left(x; Z'_{c+1}, D'_{[c+1]}\right)\middle| X_{c+1}, X'_{c+1}\right] = \mathbb{E}\left[\kappa\left(x; Z'_{c+1}, D'_{[c+1]}\right)\middle| X'_{c+1}\right] \tag{B.28}$$

Furthermore, observe the following.

$$\mathbb{E}\left[\kappa\left(x; Z'_{c+1}, D'_{[c+1]}\right)\middle| X'_{c+1}\right]\mathbb{E}\left[\kappa\left(x; Z'_{c+1}, D'_{(c+1):s}\right)\middle| X'_{c+1}\right] = \mathbb{E}\left[\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X'_{c+1}\right] \tag{B.29}$$

Thus, we can find the following.

$$\begin{aligned}
&\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_1, X'_{c+1}\right] \\
&= \mathbb{1}\left(\|X'_{c+1} - x\| \leq \|X_{c+1} - x\|\right)\left\{1 - \varphi\left(B\left(x, \|X_{c+1} - x\|\right)\right)\right\}^{s-1}\left\{1 - \varphi\left(B\left(x, \left\|X'_{c+1} - x\right\|\right)\right)\right\}^{s-c-2} \\
&\quad + \mathbb{1}\left(\|X'_{c+1} - x\| > \|X_{c+1} - x\|\right)\left\{1 - \varphi\left(B\left(x, \|X_{c+1} - x\|\right)\right)\right\}^{s-c-1}\left\{1 - \varphi\left(B\left(x, \left\|X'_{c+1} - x\right\|\right)\right)\right\}^{s-1} \\
&= \left\{1 - \varphi\left(B\left(x, \min\left(\|X_{c+1} - x\|, \left\|X'_{c+1} - x\right\|\right)\right)\right)\right\}^{s-c-1}\left\{1 - \varphi\left(B\left(x, \max\left(\|X_{c+1} - x\|, \left\|X'_{c+1} - x\right\|\right)\right)\right)\right\}^{s-1}
\end{aligned} \tag{B.30}$$

□

Lemma B.5.
For any $L^2(\mathcal{X})$ function f that is continuous at x, it holds that

$$\lim_{s\longrightarrow\infty} \underbrace{\mathbb{E}\left[f^2(X_1)(2s-c)\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\kappa\left(x;Z_1,D'_{[s]}\right)\middle|X_1\right]\right]}_{(A)} = f^2(x) \tag{B.31}$$

$$\lim_{s\longrightarrow\infty} \underbrace{\mathbb{E}\left[f(X_1)f(X'_{c+1})\frac{\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\middle|X_1,X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\right]}\right]}_{(B)} = f^2(x) \tag{B.32}$$

$$\lim_{s\longrightarrow\infty} \underbrace{\mathbb{E}\left[f(X_{c+1})f(X'_{c+1})\frac{\mathbb{E}\left[\kappa\left(x;Z_{c+1},D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\middle|X_{c+1},X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x;Z_{c+1},D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\right]}\right]}_{(C)} = f^2(x) \tag{B.33}$$

Proof of Lemma B.5.
We will largely argue along the same lines as the original proof in [6]. Thus, consider first the following inequalities.

$$\begin{aligned}
&\left|(A)-f^2(x)\right| = \left|\mathbb{E}\left[f^2(X_1)(2s-c)\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\kappa\left(x;Z_1,D'_{[s]}\right)\middle|X_1\right]\right]-f^2(x)\right| \\
&\leq \mathbb{E}\left[\left|f^2(X_1)-f^2(x)\right|(2s-c)\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\kappa\left(x;Z_1,D'_{[s]}\right)\middle|X_1\right]\right]
\end{aligned} \tag{B.34}$$

$$\begin{aligned}
&\left|(B)-f^2(x)\right| \\
&= \left|\mathbb{E}\left[f(X_1)f(X'_{c+1})\frac{\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\middle|X_1,X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\right]}\right]-f^2(x)\right| \\
&\leq \mathbb{E}\left[\left|f(X_1)f(X'_{c+1})-f^2(x)\right|\frac{\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\middle|X_1,X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\right]}\right]
\end{aligned} \tag{B.35}$$

$$\begin{aligned}
&\left|(C)-f^2(x)\right| \\
&= \left|\mathbb{E}\left[f(X_{c+1})f(X'_{c+1})\frac{\mathbb{E}\left[\kappa\left(x;Z_{c+1},D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\middle|X_{c+1},X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x;Z_{c+1},D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\right]}\right]-f^2(x)\right| \\
&\leq \mathbb{E}\left[\left|f(X_{c+1})f(X'_{c+1})-f^2(x)\right|\frac{\mathbb{E}\left[\kappa\left(x;Z_{c+1},D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\middle|X_{c+1},X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x;Z_{c+1},D_{[s]}\right)\kappa\left(x;Z'_{c+1},D'_{[s]}\right)\right]}\right]
\end{aligned} \tag{B.36}$$

Now, fix an arbitrary $\epsilon > 0$. By continuity of f at x, there exists a $\delta > 0$, such that the following holds.

$$\forall X, X' \in B(x,\delta): \quad \left|f(X)f(X') - f^2(x)\right| < \epsilon \tag{B.37}$$

We can consider decompositions of these terms in analogy to [6], i.e., by considering cases with observations lying within this sphere or outside of it, and observe the following.

$$\begin{aligned}
&\mathbb{E}\Big[\left|f^2(X_1) - f^2(x)\right| (2s-c) \\
&\quad \times \mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z_1, D'_{[s]}\right) \mathbb{1}\left(X_1 \in B(x,\delta)\right)\middle| X_1\right] \Big] \\
&\leq \epsilon \mathbb{E}\left[(2s-c)\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z_1, D'_{[s]}\right) \mathbb{1}\left(X_1 \in B(x,\delta)\right)\middle| X_1\right]\right] \\
&\leq \epsilon \mathbb{E}\left[(2s-c)\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z_1, D'_{[s]}\right)\middle| X_1\right]\right] = \epsilon
\end{aligned} \tag{B.38}$$

$$\begin{aligned}
&\mathbb{E}\Bigg[\left|f(X_1)f(X'_{c+1}) - f^2(x)\right| \mathbb{1}\left(X_1, X'_{c+1} \in B(x,\delta)\right) \\
&\qquad \times \frac{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_1, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]} \Bigg] \\
&\leq \epsilon \mathbb{E}\left[\frac{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_1, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]} \mathbb{1}\left(X_1, X'_{c+1} \in B(x,\delta)\right)\right] \\
&\leq \epsilon \mathbb{E}\left[\frac{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_1, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]}\right] = \epsilon
\end{aligned} \tag{B.39}$$

$$\begin{aligned}
&\mathbb{E}\Bigg[\left|f(X_{c+1})f(X'_{c+1}) - f^2(x)\right| \frac{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_{c+1}, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]} \\
&\quad \times \mathbb{1}\left(X_{c+1}, X'_{c+1} \in B(x,\delta)\right) \Bigg] \\
&\leq \epsilon \mathbb{E}\left[\frac{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_{c+1}, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]} \mathbb{1}\left(X_{c+1}, X'_{c+1} \in B(x,\delta)\right)\right] \\
&\leq \epsilon \mathbb{E}\left[\frac{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_{c+1}, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]}\right] = \epsilon
\end{aligned} \tag{B.40}$$

Considering next the parts of the expectation that are not covered by the previous cases, we can find the following. As in the original proof, we use the fact that if X or X' do not lie within $B(x,\delta)$, then the following holds

$$B(x,\delta) \subseteq B(x, \max\left(\|X - x\|, \|X' - x\|\right)). \tag{B.41}$$

This allows us to find the following.

$$\begin{aligned}
&\mathbb{E}\Bigg[\left|f^2(X_1) - f^2(x)\right| (2s-c) \\
&\qquad \times \mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z_1, D'_{[s]}\right) \left(1 - \mathbb{1}\left(X_1 \in B(x,\delta)\right)\right) \middle| X_1\right] \Bigg] \\
&\leq \mathbb{E}\left[\left|f^2(X_1) - f^2(x)\right| (2s-c) \left\{1 - \varphi\left(B\left(x,\delta\right)\right)\right\}^{2s-c-1} \left(1 - \mathbb{1}\left(X_1 \in B(x,\delta)\right)\right)\right] \\
&\leq (2s-c) 1 \left\{1 - \varphi\left(B\left(x,\delta\right)\right)\right\}^{2s-c-1} \mathbb{E}\left[\left|f^2(X_1) - f^2(x)\right|\right]
\end{aligned} \tag{B.42}$$

In the second case, first recall the form of the conditional expectation from Lemma B.4.

$$\begin{aligned}
&\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right) \middle| X_1, X'_{c+1}\right] \\
&= \mathbb{1}\left(\|X'_{c+1} - x\| \leq \|X_1 - x\|\right) \\
&\qquad \times \left\{1 - \varphi\left(B\left(x, \|X_1 - x\|\right)\right)\right\}^{s-1} \left\{1 - \varphi\left(B\left(x, \left\|X'_{c+1} - x\right\|\right)\right)\right\}^{s-c-1}
\end{aligned} \tag{B.43}$$

The indicator variable in this expression is only non-zero if $\max\left(\|X_1 - x\|, \|X'_{c+1} - x\|\right) = \|X_1 - x\|$. Thus, in light of the conditioning, we can observe the following.

$$B(x,\delta) \subseteq B(x, \|X_1 - x\|) \tag{B.44}$$

Thus, we can make the following observation.

$$\begin{aligned}
&\mathbb{E}\Bigg[\left|f(X_1) f(X'_{c+1}) - f^2(x)\right| \frac{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right) \middle| X_1, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]} \\
&\qquad \left(1 - \mathbb{1}\left(X_1, X'_{c+1} \in B(x,\delta)\right)\right) \Bigg] \\
&\overset{\text{(Lem B.3)}}{\leq} (2s-c)^2 \mathbb{E}\Big[\left|f(X_1) f(X'_{c+1}) - f^2(x)\right| \\
&\qquad \mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right) \middle| X_1, X'_{c+1}\right] \left(1 - \mathbb{1}\left(X_1, X'_{c+1} \in B(x,\delta)\right)\right) \Big] \\
&\overset{\text{(Lem B.4)}}{=} (2s-c)^2 \mathbb{E}\Big[\left|f(X_1) f(X'_{c+1}) - f^2(x)\right| \mathbb{1}\left(\|X'_{c+1} - x\| \leq \|X_1 - x\|\right) \\
&\qquad \times \left\{1 - \varphi\left(B\left(x, \|X_1 - x\|\right)\right)\right\}^{s-1} \left\{1 - \varphi\left(B\left(x, \left\|X'_{c+1} - x\right\|\right)\right)\right\}^{s-c-1} \\
&\qquad \times \left(1 - \mathbb{1}\left(X_1, X'_{c+1} \in B(x,\delta)\right)\right) \Big] \\
&\leq (2s-c)^2 \left\{1 - \varphi\left(B\left(x,\delta\right)\right)\right\}^{s-1} \\
&\qquad \times \mathbb{E}\Big[\left|f(X_1) f(X'_{c+1}) - f^2(x)\right| \mathbb{1}\left(\delta < \|X'_{c+1} - x\| \leq \|X_1 - x\|\right) \Big] \\
&\leq (2s-c)^2 \left\{1 - \varphi\left(B\left(x,\delta\right)\right)\right\}^{s-1} \mathbb{E}\left[\left|f(X_1) f(X'_{c+1}) - f^2(x)\right|\right]
\end{aligned} \tag{B.45}$$

Similarly, considering the third case, we observe the following.

$$
\begin{aligned}
&\mathbb{E}\Bigg[\left| f(X_{c+1}) f(X'_{c+1}) - f^2(x) \right| \frac{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right) \middle| X_{c+1}, X'_{c+1} \right]}{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right) \right]} \left(1 - \mathbb{1}\left(X_{c+1}, X'_{c+1} \in B(x,\delta)\right)\right) \Bigg] \\
&\overset{\text{(Lem B.3)}}{\leq} (2s-c)^2 \mathbb{E}\Big[\left| f(X_{c+1}) f(X'_{c+1}) - f^2(x) \right| \mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right) \middle| X_{c+1}, X'_{c+1} \right] \left(1 - \mathbb{1}\left(X_{c+1}, X'_{c+1} \in B(x,\delta)\right)\right) \Big] \\
&\overset{\text{(Lem B.4)}}{=} (2s-c)^2 \mathbb{E}\Big[\left| f(X_{c+1}) f(X'_{c+1}) - f^2(x) \right| \left\{1 - \varphi\left(B\left(x, \min\left(\|X_{c+1} - x\|, \left\|X'_{c+1} - x\right\|\right)\right)\right)\right\}^{s-c-1} \\
&\qquad \times \left\{1 - \varphi\left(B\left(x, \max\left(\|X_{c+1} - x\|, \left\|X'_{c+1} - x\right\|\right)\right)\right)\right\}^{s-1} \left(1 - \mathbb{1}\left(X_{c+1}, X'_{c+1} \in B(x,\delta)\right)\right) \Big] \\
&\leq (2s-c)^2 \mathbb{E}\Big[\left| f(X_{c+1}) f(X'_{c+1}) - f^2(x) \right| \left\{1 - \varphi\left(B\left(x, \min\left(\|X_{c+1} - x\|, \left\|X'_{c+1} - x\right\|\right)\right)\right)\right\}^{s-c-1} \\
&\qquad \times \left\{1 - \varphi\left(B\left(x,\delta\right)\right)\right\}^{s-1} \left(1 - \mathbb{1}\left(X_{c+1}, X'_{c+1} \in B(x,\delta)\right)\right) \Big] \\
&\leq (2s-c)^2 \left\{1 - \varphi\left(B\left(x,\delta\right)\right)\right\}^{s-1} \mathbb{E}\Big[\left| f(X_{c+1}) f(X'_{c+1}) - f^2(x) \right| \Big]
\end{aligned}
\tag{B.46}
$$

Concerning the resulting terms in these three expressions, we can then make the following observations.

$$
\mathbb{E}\Big[\left| f^2(X) - f^2(x) \right| \Big] \leq \mathbb{E}\Big[f^2(X) \Big] + f^2(x) = \|f\|_{L_2}^2 + f^2(x) \tag{B.47}
$$

$$
\begin{aligned}
\mathbb{E}\Big[\left| f(X) f(X') - f^2(x) \right| \Big] &\leq \mathbb{E}\Big[\left| f(X) f(X') \right| \Big] + f^2(x) \\
&\leq \mathbb{E}\Big[\left| f(X) \right| \left| f(X') \right| \Big] + f^2(x) = \|f\|_{L^1}^2 + f^2(x)
\end{aligned}
\tag{B.48}
$$

As f is an $L^2(\mathcal{X})$ function on a bounded domain, observe that $\|f\|_{L^1}$ is finite. Thus, we can find the following.

$$
\mathbb{E}\left[\left| f^2(X_1) - f^2(x) \right| (2s-c) \mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z_1, D'_{[s]}\right) \left(1 - \mathbb{1}\left(X_1 \in B(x,\delta)\right)\right) \middle| X_1 \right] \right] \longrightarrow 0 \quad \text{as} \quad s \longrightarrow \infty \tag{B.49}
$$

$$\mathbb{E}\left[\left|f(X_1)f(X'_{c+1}) - f^2(x)\right| \frac{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_1, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]}\left(1 - \mathbb{1}\left(X_1, X'_{c+1} \in B(x,\delta)\right)\right)\right] \longrightarrow 0 \quad \text{as} \quad s \longrightarrow \infty \tag{B.50}$$

$$\mathbb{E}\left[\left|f(X_{c+1})f(X'_{c+1}) - f^2(x)\right| \frac{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_{c+1}, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]}\left(1 - \mathbb{1}\left(X_{c+1}, X'_{c+1} \in B(x,\delta)\right)\right)\right] \longrightarrow 0 \quad \text{as} \quad s \longrightarrow \infty \tag{B.51}$$

Combining these findings, for large enough s we can bound the terms $\|(A) - f^2(x)\|$, $\|(B) - f^2(x)\|$, and $\|(C) - f^2(x)\|$ by 2ϵ, respectively. As ϵ was arbitrary this concludes the proof.

□

Lemma B.6.

The following inequalities hold.

$$\begin{aligned}&\forall i \in [c]\ \forall j \in \{c+1,\dots,s\}:\\ &\mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z'_j, D'_{[s]}\right)\right] \leq \frac{s}{(2s-c)(2s-c-1)(c+1)}\end{aligned} \tag{B.52}$$

$$\begin{aligned}&\forall i, j \in \{c+1,\dots,s\}:\\ &\mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z'_j, D'_{[s]}\right)\right] \leq \frac{2(s-c)}{(2s-c)^2(2s-c-1)}\end{aligned} \tag{B.53}$$

Proof of Lemma B.6.

Recall the results of Lemma B.3 and make the following observations.

$$\begin{aligned}&\mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z'_j, D'_{[s]}\right)\right]\\ &= \frac{1}{(2s-c)(2s-c-1)}\sum_{i=0}^{s-c-1}\binom{s-c-1}{i}\binom{2s-c-2}{i}^{-1}\\ &\leq \frac{1}{(2s-c)(2s-c-1)}\sum_{i=0}^{s-c-1}\frac{(s-c-1)^i}{i!}\frac{i!}{(2s-c-1-i)^i}\\ &= \frac{1}{(2s-c)(2s-c-1)}\sum_{i=0}^{s-c-1}\left(\frac{s-c-1}{2s-c-1-i}\right)^i\\ &\leq \frac{1}{(2s-c)(2s-c-1)}\sum_{i=0}^{s-c-1}\left(\frac{s-c-1}{2s-c-1-i}\right)^i\\ &\leq \frac{1}{(2s-c)(2s-c-1)}\sum_{i=0}^{s-c-1}\left(\frac{s-c-1}{s}\right)^i\\ &\leq \frac{1}{(2s-c)(2s-c-1)}\sum_{i=0}^{\infty}\left(\frac{s-c-1}{s}\right)^i = \frac{s}{(2s-c)(2s-c-1)(c+1)}\end{aligned} \tag{B.54}$$

Similarly, for the second case, we can make the following observation.

$$\begin{aligned}
&\mathbb{E}\left[\kappa\left(x; Z_i, D_{[s]}\right)\kappa\left(x; Z'_j, D'_{[s]}\right)\right] \\
&= \frac{2}{(2s-c)(2s-c-1)}\sum_{i=0}^{s-c-1}\binom{s-c-1}{i}\binom{2s-c-1}{s-1+i}^{-1} \\
&= \frac{2}{(2s-c)(2s-c-1)}\sum_{i=0}^{s-c-1}\binom{s-c-1}{i}\binom{2s-c-1}{s-c-i}^{-1} \\
&= \frac{2}{(2s-c)(2s-c-1)}\sum_{i=0}^{s-c-1}\frac{(s-c-1)!}{(s-c-1-i)!i!}\frac{(s-1+i)!(s-c-i)!}{(2s-c-1)!} \\
&= \frac{2}{(2s-c)(2s-c-1)}\frac{(s-c-1)!(s-1)!}{(2s-c-1)!}\sum_{i=0}^{s-c-1}(s-c-i)\binom{s-1+i}{i} \\
&\leq \frac{2}{(2s-c)(2s-c-1)}\frac{(s-c)!(s-1)!}{(2s-c-1)!}\sum_{i=0}^{s-c-1}\binom{s-1+i}{i} \\
&= \frac{2}{(2s-c)(2s-c-1)}\binom{2s-c}{s-c}^{-1}\binom{2s-c-1}{s-c-1} = \frac{2(s-c)}{(2s-c)^2(2s-c-1)}
\end{aligned} \tag{B.55}$$

□

B.3. (T)DNN Kernel Expectations

As part of deriving consistency results for the variance estimators under consideration, we need to do a careful analysis of the kernel of the DNN and TDNN estimators. In this section of the appendix, we will thus derive the expectations of the kernel and its corresponding Hájek projection. First, we start with the nonparametric regression setup.

Lemma B.7 (NPR - DNN Kernel Expectation)**.**
Let x denote a point of interest. Then

$$\mathbb{E}\left[h_s\left(x; D_{[s]}\right)\right] = \mathbb{E}\left[Y_1 s\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\middle|X_1\right]\right] \longrightarrow \mu(x) \quad as \quad s \longrightarrow \infty \tag{B.56}$$

Proof of Lemma B.7. This result follows immediately from Lemma B.2 and the following observation.

$$\begin{aligned}
&\mathbb{E}\left[Y_1 s\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\middle|X_1\right]\right] = \mathbb{E}\left[\left(\mu(X_1)+\varepsilon_1\right)s\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\middle|X_1\right]\right] \\
&= \mathbb{E}\left[\left(\mu(X_1)+\mathbb{E}\left[\varepsilon_1|X_1\right]\right)s\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\middle|X_1\right]\right] \\
&= \mathbb{E}\left[\mu(X_1)\, s\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\middle|X_1\right]\right] \\
&\overset{\text{(Lem B.2)}}{\longrightarrow} \mu(x) \quad \text{as} \quad s \longrightarrow \infty
\end{aligned} \tag{B.57}$$

□

Lemma B.8 (NPR - DNN Hájek Kernel Expectation)**.**
Let $z_1 = (x_1, y_1)$ denote a specific realization of Z and x denote a point of interest. Then

$$\psi_s^1(x; z_1) = \varepsilon_1\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\middle|X_1 = x_1\right] + \mathbb{E}\left[\sum_{i=2}^{s}\kappa\left(x; Z_i, D_{[s]}\right)\mu(X_i)\middle|X_1 = x_1\right] \tag{B.58}$$

Proof of Lemma B.8.

$$\begin{aligned}\psi_s^1(x;z_1) &= \mathbb{E}\left[h_s\left(x;D_{[s]}\right)\middle|Z_1=z_1\right] = \mathbb{E}\left[\sum_{i=1}^{s}\kappa\left(x;Z_i,D_{[s]}\right)Y_i\middle|Z_1=z_1\right]\\ &= \mathbb{E}\left[\left(\mu(x_1)+\varepsilon_1\right)\kappa\left(x;Z_1,D_{[s]}\right)+\sum_{i=2}^{s}\kappa\left(x;Z_i,D_{[s]}\right)\mu(X_i)\middle|Z_1=z_1\right] \qquad \text{(B.59)}\\ &= \varepsilon_1\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\middle|X_1=x_1\right]+\mathbb{E}\left[\sum_{i=2}^{s}\kappa\left(x;Z_i,D_{[s]}\right)\mu(X_i)\middle|X_1=x_1\right]\end{aligned}$$

□

B.4. DNN Kernel Variances & Covariances

Similar to the previous section of proofs, we will continue by analyzing the variances and covariances of the kernels under consideration. These results will play an important role in the derivation of consistency properties for the variance estimators. Similar to the previous part, we will first consider the nonparametric regression setup and then proceed to the conditional average treatment effect setup.

Lemma B.9 (Adapted from [6])**.**
Let $D_{[s]}=\{Z_1,\ldots,Z_s\}$ be a vector of i.i.d. random variables drawn from P. Furthermore, let

$$\Omega_s(x) = \mathbb{E}\left[h_s^2\left(x;D_{[s]}\right)\right]. \qquad \text{(B.60)}$$

Then,

$$\Omega_s(x) = \mathbb{E}\left[\left(\mu(X_1)+\varepsilon_1\right)^2 s\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\middle|X_1\right]\right] \lesssim \mu^2(x)+\overline{\sigma}_\varepsilon^2+o(1) \qquad \text{(B.61)}$$

Proof of Lemma B.9.
This result follows immediately from Lemma B.2 and the following observation.

$$\begin{aligned}\Omega_s(x) &= \mathbb{E}\left[h_s^2\left(x;D_{[s]}\right)\right] = \mathbb{E}\left[\left(\sum_{i=1}^{s}\kappa\left(x;Z_i,D_{[s]}\right)Y_i\right)^2\right]\\ &= \mathbb{E}\left[\sum_{i=1}^{s}\sum_{j=1}^{s}\left(\kappa\left(x;Z_i,D_{[s]}\right)\kappa\left(x;Z_j,D_{[s]}\right)Y_iY_j\right)\right] = \mathbb{E}\left[s\kappa\left(x;Z_1,D_{[s]}\right)Y_1^2\right]\\ &= \mathbb{E}\left[Y_1^2 s\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\middle|X_1\right]\right] = \mathbb{E}\left[\left(\mu(X_1)+\varepsilon_1\right)^2 s\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\middle|X_1\right]\right] \qquad \text{(B.62)}\\ &= \mathbb{E}\left[\left(\mu^2(X_1)+2\mu(X_1)\varepsilon_1+\varepsilon_1^2\right)s\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\middle|X_1\right]\right]\\ &= \mathbb{E}\left[\left(\mu^2(X_1)+2\mu(X_1)\mathbb{E}\left[\varepsilon_1|X_1\right]+\mathbb{E}\left[\varepsilon_1^2\middle|X_1\right]\right)s\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\middle|X_1\right]\right]\\ &= \mathbb{E}\left[\left(\mu^2(X_1)+\sigma_\varepsilon^2(X_1)\right)s\mathbb{E}\left[\kappa\left(x;Z_1,D_{[s]}\right)\middle|X_1\right)\right]\right]\\ &\overset{\text{(Lem B.2)}}{\longrightarrow} \mu^2(x)+\sigma_\varepsilon^2(x) \quad \text{as} \quad s\longrightarrow\infty\end{aligned}$$

Furthermore, we have the following inequality.

$$\mu^2(x) + \sigma_\varepsilon^2(x) \leq \mu^2(x) + \overline{\sigma}_\varepsilon^2 \tag{B.63}$$

Thus, we obtain the desired result. □

Lemma B.10.
Let $D_{[s]} = \{Z_1, \ldots, Z_s\}$ be a vector of i.i.d. random variables drawn from P. Let $D'_{[s]} = \{Z_1, \ldots, Z_c, Z'_{c+1}, \ldots, Z'_s\}$ where $Z'_{c+1}, \ldots, Z'_s$ are i.i.d. draws from P that are independent of $D_{[s]}$. Furthermore, let

$$\Omega_s^c(x) = \mathbb{E}\left[h_s\left(x; D_{[s]}\right) h_s\left(x; D'_{[s]}\right)\right]. \tag{B.64}$$

Then,

$$\Omega_s^c(x) \lesssim \mu^2(x) + \overline{\sigma}_\varepsilon^2 + o(1) \tag{B.65}$$

Proof of Lemma B.10.

$$\begin{aligned}
\Omega_s^c(x) &= \mathbb{E}\left[h_s\left(x; D_{[s]}\right) h_s\left(x; D'_{[s]}\right)\right] \\
&= \mathbb{E}\left[\left(\sum_{i=1}^{s} \kappa\left(x; Z_i, D_{[s]}\right) Y_i\right)\left(\sum_{j=1}^{c} \kappa\left(x; Z_j, D'_{[s]}\right) Y_j + \sum_{j=c+1}^{s} \kappa\left(x; Z'_j, D'_{[s]}\right) Y'_j\right)\right] \\
&= \underbrace{\mathbb{E}\left[c\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z_1, D'_{[s]}\right) Y_1^2\right]}_{(A)} \\
&\quad + \underbrace{2\,\mathbb{E}\left[c(s-c)\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right) Y_1 Y'_{c+1}\right]}_{(B)} \\
&\quad + \underbrace{\mathbb{E}\left[(s-c)^2\kappa\left(x; Z_{c+1}, D_{[s]}\right) \kappa\left(x; Z'_{c+1}, D'_{[s]}\right) Y_{c+1} Y'_{c+1}\right]}_{(C)}
\end{aligned} \tag{B.66}$$

Starting from this decomposition, we will analyze the terms one by one using Lemma B.5.

$$\begin{aligned}
(A) &= \mathbb{E}\left[c\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z_1, D'_{[s]}\right) Y_1^2\right] \\
&= \frac{c}{2s-c}\mathbb{E}\left[(\mu(X_1) + \varepsilon_1)^2 (2s-c)\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z_1, D'_{[s]}\right)\middle| X_1\right]\right] \\
&= \frac{c}{2s-c}\mathbb{E}\left[\left(\mu^2(X_1) + \sigma^2(X_1)\right)(2s-c)\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right) \kappa\left(x; Z_1, D'_{[s]}\right)\middle| X_1\right]\right] \\
&\overset{\text{(Lem B.5)}}{\lesssim} \frac{c}{2s-c}\left(\mu^2(x) + \sigma_\varepsilon(x)\right) + o(1)
\end{aligned} \tag{B.67}$$

Similarly, we can find the following.

$$
\begin{aligned}
(B) &= \mathbb{E}\left[c(s-c)\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right) Y_1 Y'_{c+1}\right] \\
&\overset{\text{(Lem B.6)}}{\leq} \frac{c(s-c)s}{(2s-c)(2s-c-1)(c+1)}\mathbb{E}\Bigg[\left(\mu(X_1)+\varepsilon_1\right)\left(\mu(X'_{c+1})+\varepsilon'_{c+1}\right) \\
&\qquad \times \frac{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_1, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]}\Bigg] \\
&\leq \frac{(s-c)s}{(2s-c)(2s-c-1)} \\
&\qquad \times \mathbb{E}\left[\mu(X_1)\mu(X'_{c+1})\frac{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_1, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_1, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]}\right] \\
&\overset{\text{(Lem B.5)}}{\lesssim} \frac{(s-c)s}{(2s-c)(2s-c-1)}\mu^2(x) + o(1)
\end{aligned}
\tag{B.68}
$$

The third term can be asymptotically bounded in the following way.

$$
\begin{aligned}
(C) &= \mathbb{E}\left[(s-c)^2\kappa\left(x; Z_{c+1}, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right) Y_{c+1} Y'_{c+1}\right] \\
&\overset{\text{(Lem B.6)}}{\leq} \frac{2(s-c)^3}{(2s-c)^2(2s-c-1)} \\
&\qquad \mathbb{E}\left[\mu(X_{c+1})\mu(X'_{c+1})\frac{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\middle| X_{c+1}, X'_{c+1}\right]}{\mathbb{E}\left[\kappa\left(x; Z_{c+1}, D_{[s]}\right)\kappa\left(x; Z'_{c+1}, D'_{[s]}\right)\right]}\right] \\
&\overset{\text{(Lem B.5)}}{\lesssim} \frac{2(s-c)}{(2s-c-1)}\mu^2(x) + o(1)
\end{aligned}
\tag{B.69}
$$

The result of Lemma B.10 follows immediately by summing up the asymptotic bounds for the individual terms. □

B.5. TDNN Kernel Variances & Covariances

Now we will introduce a variation of the setup we have considered so far. Consider a sample size n, sequences of subsampling scales $0 < s_1 < s_2 < n$ growing with n and satisfying Assumption 5, and c such that $0 < c \leq s_1 \leq n$. Let $D_{[s_2]} = \left\{Z_1, Z_2, \ldots, Z_c, Z_{c+1}, \ldots Z_{s_2}\right\}$ be an i.i.d. data set drawn from P as described in Setup 3. Let $D'_{[s_1]} = \left\{Z_1, Z_2, \ldots, Z_c, Z'_{c+1}, \ldots Z'_{s_1}\right\}$ be a second data set that shares the first c observations with $D_{[s_2]}$. The remaining $s_1 - c$ observations of $D'_{[s_1]}$, i.e. $\left\{Z'_{c+1}, \ldots Z'_{s_1}\right\}$, are i.i.d. draws from P that are independent of $D_{[s_2]}$.

Lemma B.11.
Let

$$
\Upsilon_{s_1,s_2}(x) = \mathbb{E}\left[h_{s_1}\left(x; D_{[s_1]}\right) h_{s_2}\left(x; D_{[s_2]}\right)\right]. \tag{B.70}
$$

Then,

$$\Upsilon_{s_1,s_2}(x) \lesssim \mu^2(x) + \overline{\sigma}_{\varepsilon}^2 + o(1) \tag{B.71}$$

Proof of Lemma B.11.

$$\begin{aligned}
&\Upsilon_{s_1,s_2}(x) = \mathbb{E}\left[h_{s_1}\left(x; D_{[s_1]}\right) h_{s_2}\left(x; D_{[s_2]}\right)\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{s_1} \kappa\left(x; Z_i, D_{[s_1]}\right) Y_i \left(\sum_{j=1}^{s_1} \kappa\left(x; Z_j, D_{[s_2]}\right) Y_j + \sum_{j=s_1+1}^{s_2} \kappa\left(x; Z_j, D_{[s_2]}\right) Y_j\right)\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{s_1} \kappa\left(x; Z_i, D_{[s_2]}\right) Y_i^2\right] + \mathbb{E}\left[\sum_{i=1}^{s_1}\sum_{j=s_1+1}^{s_2} \kappa\left(x; Z_i, D_{[s_1]}\right) \kappa\left(x; Z_j, D_{[s_2]}\right) Y_i Y_j\right] \\
&= \mathbb{E}\left[Y_1^2\, s_1\, \kappa\left(x; Z_1, D_{[s_2]}\right)\right] + \mathbb{E}\left[Y_1 Y_{s_2}\, s_1(s_2 - s_1)\, \kappa\left(x; Z_1, D_{[s_1]}\right) \kappa\left(x; Z_{s_2}, D_{[s_2]}\right)\right] \\
&= \mathbb{E}\left[\left(\mu^2(X_1) + \sigma_{\varepsilon}^2(X_1)\right) s_1\, \kappa\left(x; Z_1, D_{[s_2]}\right)\right] \\
&\quad + \mathbb{E}\left[\mu(X_1)\mu(X_{s_2})\, s_1(s_2 - s_1)\, \kappa\left(x; Z_1, D_{[s_1]}\right) \kappa\left(x; Z_{s_2}, D_{[s_2]}\right)\right] \\
&= \frac{s_1}{s_2}\mathbb{E}\left[\left(\mu^2(X_1) + \sigma_{\varepsilon}^2(X_1)\right) s_1\, \kappa\left(x; Z_1, D_{[s_2]}\right)\right] \\
&\quad + \frac{s_2 - s_1}{s_2}\mathbb{E}\left[\mu(X_1)\mu(X_{s_2})\, s_1 s_2\, \kappa\left(x; Z_1, D_{[s_1]}\right) \kappa\left(x; Z_{s_2}, D_{[s_2]}\right)\right] \\
&\leq \frac{s_1}{s_2}\mathbb{E}\left[\left(\mu^2(X_1) + \sigma_{\varepsilon}^2(X_1)\right) s_2\, \kappa\left(x; Z_1, D_{[s_2]}\right)\right] \\
&\quad + \frac{s_2 - s_1}{s_2}\mathbb{E}\left[|\mu(X_1)|\, s_1\, \kappa\left(x; Z_1, D_{[s_1]}\right)\right] \mathbb{E}\left[|\mu(X_{s_2})|\, s_2\, \kappa\left(x; Z_{s_2}, D_{[s_2]}\right)\right] \\
&\lesssim \mu^2(x) + \sigma_{\varepsilon}^2(x) + o(1) \leq \mu^2(x) + \overline{\sigma}_{\varepsilon}^2 + o(1).
\end{aligned} \tag{B.72}$$

□

Lemma B.12.
Let

$$\Upsilon_{s_1,s_2}^{c}(x) = \mathbb{E}\left[h_{s_1}\left(x; D'_{[s_1]}\right) h_{s_2}\left(x; D_{[s_2]}\right)\right]. \tag{B.73}$$

Then,

$$\Upsilon_{s_1,s_2}^{c}(x) \lesssim \frac{cs_2 - c^2 + s_1 s_2}{s_1 s_2}\mu^2(x) + (c/s_1)\overline{\sigma}_{\varepsilon}^2 + o(1) \tag{B.74}$$

and thus

$$\Upsilon_{s_1,s_2}^{c}(x) \lesssim \mu^2(x) + o(1) \tag{B.75}$$

Proof of Lemma B.12.

$$
\begin{aligned}
\Upsilon^{c}_{s_1,s_2}(x) &= \mathbb{E}\left[h_{s_1}\left(x; D'_{[s_1]}\right) h_{s_2}\left(x; D_{[s_2]}\right)\right] \\
&= \mathbb{E}\Bigg[\left(\sum_{i=1}^{c} \kappa\left(x; Z_i, D'_{[s_1]}\right) Y_i + \sum_{i=c+1}^{s_1} \kappa\left(x; Z'_i, D'_{[s_1]}\right) Y'_i\right) \\
&\quad \left(\sum_{j=1}^{c} \kappa\left(x; Z_j, D_{[s_2]}\right) Y_j + \sum_{j=c+1}^{s_2} \kappa\left(x; Z_j, D_{[s_2]}\right) Y_j\right)\Bigg] \\
&= \underbrace{\mathbb{E}\left[\sum_{i=1}^{c}\sum_{j=1}^{c} \kappa\left(x; Z_i, D'_{[s_1]}\right) \kappa\left(x; Z_j, D_{[s_2]}\right) Y_i Y_j\right]}_{(A)} \\
&\quad + \underbrace{\mathbb{E}\left[\sum_{i=1}^{c}\sum_{j=c+1}^{s_2} Y_i Y_j \kappa\left(x; Z_i, D'_{[s_1]}\right) \kappa\left(x; Z_j, D_{[s_2]}\right)\right]}_{(B)} \\
&\quad + \underbrace{\mathbb{E}\left[\sum_{i=c+1}^{s_1}\sum_{j=1}^{c} \kappa\left(x; Z'_i, D'_{[s_1]}\right) \kappa\left(x; Z_j, D_{[s_2]}\right) Y'_i Y_j\right]}_{(C)} \\
&\quad + \underbrace{\mathbb{E}\left[\sum_{i=c+1}^{s_1}\sum_{j=c+1}^{s_2} \kappa\left(x; Z'_i, D'_{[s_1]}\right) \kappa\left(x; Z_j, D_{[s_2]}\right) Y'_i Y_j\right]}_{(D)}
\end{aligned}
\tag{B.76}
$$

Again, we have four terms to analyze individually.

$$
\begin{aligned}
(A) &= \mathbb{E}\left[\sum_{i=1}^{c}\sum_{j=1}^{c} \kappa\left(x; Z_i, D'_{[s_1]}\right) \kappa\left(x; Z_j, D_{[s_2]}\right) Y_i Y_j\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{c} Y_i^2 \kappa\left(x; Z_i, D'_{[s_1]}\right) \kappa\left(x; Z_i, D_{[s_2]}\right)\right] \\
&= \mathbb{E}\left[Y_1^2\, c\, \kappa\left(x; Z_1, D'_{[s_1]}\right) \kappa\left(x; Z_1, D_{[s_2]}\right)\right] \\
&= \mathbb{E}\left[\left(\mu^2(X_1) + \sigma^2_{\varepsilon}(X_1)\right)\, c\, \kappa\left(x; Z_1, D_{[s_2]}\right) \kappa\left(x; Z_1, D'_{c+1:s_1}\right)\right] \\
&\leq \mathbb{E}\left[\left(\mu^2(X_1) + \sigma^2_{\varepsilon}(X_1)\right)\, c\, \kappa\left(x; Z_1, D_{[s_2]}\right)\right] \\
&= \frac{c}{s_1}\mathbb{E}\left[\left(\mu^2(X_1) + \sigma^2_{\varepsilon}(X_1)\right)\, s_1\, \kappa\left(x; Z_1, D_{[s_2]}\right)\right] \\
&\lesssim (c/s_1)(\mu^2(x) + \sigma^2_{\varepsilon}(x)) + o(1) \leq (c/s_1)(\mu^2(x) + \overline{\sigma}^2_{\varepsilon}) + o(1)
\end{aligned}
\tag{B.77}
$$

Considering the second term, we find the following.

$$\begin{aligned}
(B) &= \mathbb{E}\left[\sum_{i=1}^{c}\sum_{j=c+1}^{s_2} Y_i Y_j \kappa\left(x; Z_i, D'_{[s_1]}\right)\kappa\left(x; Z_j, D_{[s_2]}\right)\right] \\
&= \mathbb{E}\left[c(s_2-c)\, Y_1 Y_{s_1} \kappa\left(x; Z_1, D'_{[s_1]}\right)\kappa\left(x; Z_{s_2}, D_{[s_2]}\right)\right] \\
&= \frac{c(s_2-c)}{s_1 s_2}\mathbb{E}\left[Y_1 Y_{s_2}\, s_1 s_2\, \kappa\left(x; Z_1, D'_{[s_1]}\right)\kappa\left(x; Z_{s_2}, D_{[s_2]}\right)\right] \\
&\leq \frac{c(s_2-c)}{s_1 s_2}\mathbb{E}\left[|\mu(X_1)|\, s_1\, \kappa\left(x; Z_1, D'_{[s_1]}\right)\right]\mathbb{E}\left[|\mu(X_{s_2})|\, s_2\, \kappa\left(x; Z_{s_2}, D_{[s_2]}\right)\right] \\
&\lesssim \frac{c(s_2-c)}{s_1 s_2}\mu^2(x) + o(1)
\end{aligned} \tag{B.78}$$

Similarly, by simplifying the third term, we find the following.

$$\begin{aligned}
(C) &= \mathbb{E}\left[\sum_{i=c+1}^{s_1}\sum_{j=1}^{c} \kappa\left(x; Z'_i, D'_{[s_1]}\right)\kappa\left(x; Z_j, D_{[s_2]}\right) Y'_i Y_j\right] \\
&= \mathbb{E}\left[Y'_{s_1} Y_1\, (s_1-c)c\, \kappa\left(x; Z'_{s_1}, D'_{[s_1]}\right)\kappa\left(x; Z_1, D_{[s_2]}\right)\right] \\
&= \frac{(s_1-c)c}{s_1 s_2}\mathbb{E}\left[\mu(X'_{s_1})\mu(X_1)\, s_1 s_2\, \kappa\left(x; Z'_{s_1}, D'_{[s_1]}\right)\kappa\left(x; Z_1, D_{[s_2]}\right)\right] \\
&\leq \frac{(s_1-c)c}{s_1 s_2}\mathbb{E}\left[|\mu(X'_{s_1})|\, s_1\, \kappa\left(x; Z'_{s_1}, D'_{[s_1]}\right)\right]\mathbb{E}\left[|\mu(X_1)|\, s_2\, \kappa\left(x; Z_1, D_{[s_2]}\right)\right] \\
&\lesssim \frac{(s_1-c)c}{s_1 s_2}\mu^2(x) + o(1)
\end{aligned} \tag{B.79}$$

Lastly, concerning the fourth term, observe the following.

$$\begin{aligned}
(D) &= \mathbb{E}\left[\sum_{i=c+1}^{s_1}\sum_{j=c+1}^{s_2} \kappa\left(x; Z'_i, D'_{[s_1]}\right)\kappa\left(x; Z_j, D_{[s_2]}\right) Y'_i Y_j\right] \\
&= \mathbb{E}\left[\mu(X'_{s_1})\mu(X_{s_2})\, (s_1-c)(s_2-c)\, \kappa\left(x; Z'_{s_1}, D'_{[s_1]}\right)\kappa\left(x; Z_{s_2}, D_{[s_2]}\right)\right] \\
&= \frac{(s_1-c)(s_2-c)}{s_1 s_2}\mathbb{E}\left[\mu(X'_{s_1})\mu(X_{s_2})\, s_1 s_2\, \kappa\left(x; Z'_{s_1}, D'_{[s_1]}\right)\kappa\left(x; Z_{s_2}, D_{[s_2]}\right)\right] \\
&\leq \frac{(s_1-c)(s_2-c)}{s_1 s_2} \\
&\quad \times \mathbb{E}\left[|\mu(X'_{s_1})|\, s_1\, \kappa\left(x; Z'_{s_1}, D'_{[s_1]}\right)\right]\mathbb{E}\left[|\mu(X_{s_2})|\, s_2\, \kappa\left(x; Z_{s_2}, D_{[s_2]}\right)\right] \\
&\lesssim \frac{(s_1-c)(s_2-c)}{s_1 s_2}\mu^2(x) + o(1)
\end{aligned} \tag{B.80}$$

□

Lemma B.13 (Kernel Variance of the TDNN Kernel)**.**
For the kernel of the TDNN estimator with subsampling scales s_1 and s_2, it holds that

$$\zeta_{s_1,s_2}^{s_2}(x) \lesssim \mu^2(x) + \overline{\sigma}_\varepsilon^2 + o(1) \tag{B.81}$$

Proof of Lemma B.13. Consider first the following decomposition.

$$\begin{aligned}
&\zeta_{s_1,s_2}^{s_2}(x) = \operatorname{Var}\left(h_{s_1,s_2}\left(x; D_{[s_2]}\right)\right) \\
&\leq \mathbb{E}\left[h_{s_1,s_2}^2\left(x; D_{[s_2]}\right)\right] = \mathbb{E}\left[\left(w_1^* \tilde{\mu}_{s_1}\left(x; D_{[s_2]}\right) + w_2^* h_{s_2}\left(x; D_{[s_2]}\right)\right)^2\right] \\
&= \left(w_1^*\right)^2 \mathbb{E}\left[\tilde{\mu}_{s_1}^2\left(x; D_{[s_2]}\right)\right] + 2w_1^* w_2^* \mathbb{E}\left[\tilde{\mu}_{s_1}\left(x; D_{[s_2]}\right) h_{s_2}\left(x; D_{[s_2]}\right)\right] + \left(w_2^*\right)^2 \Omega_{s_2}
\end{aligned} \tag{B.82}$$

Then, observe the following.

$$\begin{aligned}
&\mathbb{E}\left[\tilde{\mu}_{s_1}^2\left(x; D_{[s_2]}\right)\right] = \mathbb{E}\left[\left(\binom{s_2}{s_1}^{-1} \sum_{\ell \in L_{s_2,s_1}} h_{s_1}\left(x; D_\ell\right)\right)^2\right] \\
&= \binom{s_2}{s_1}^{-2} \mathbb{E}\left[\sum_{\iota,\iota' \in L_{s_2,s_1}} h_{s_1}\left(x; D_\iota\right) h_{s_1}\left(x; D_{\iota'}\right)\right] \\
&= \binom{s_2}{s_1}^{-2} \sum_{c=0}^{s_1} \binom{s_2}{s_1}\binom{s_1}{c}\binom{s_2 - s_1}{s_1 - c} \Omega_{s_1}^c = \binom{s_2}{s_1}^{-1} \sum_{c=0}^{s_1} \binom{s_1}{c}\binom{s_2 - s_1}{s_1 - c} \Omega_{s_1}^c \\
&\lesssim \Omega_{s_1} \lesssim \mu(x)^2 + \overline{\sigma}_\varepsilon^2 + o(1) \quad \text{as} \quad s \longrightarrow \infty
\end{aligned} \tag{B.83}$$

Recall that by Lemma B.9, we have the following.

$$\Omega_{s_2} \lesssim \mu(x)^2 + \overline{\sigma}_\varepsilon^2 + o(1) \quad \text{as} \quad s \longrightarrow \infty \tag{B.84}$$

Lastly, consider the following.

$$\begin{aligned}
\mathbb{E}\left[\tilde{\mu}_{s_1}\left(x; D_{[s_2]}\right) h_{s_2}\left(x; D_{[s_2]}\right)\right] &= \mathbb{E}\left[\binom{s_2}{s_1}^{-1} \sum_{\ell \in L_{s_2,s_1}} h_{s_1}\left(x; D_\ell\right) h_{s_2}\left(x; D_{[s_2]}\right)\right] \\
&= \mathbb{E}\left[h_{s_1}\left(x; D_{[s_1]}\right) h_{s_2}\left(x; D_{[s_2]}\right)\right] = \Upsilon_{s_1,s_2}(x)
\end{aligned} \tag{B.85}$$

Thus, we find the following.

$$\begin{aligned}
\zeta_{s_1,s_2}^{s_2}(x) &\lesssim \left(w_1^*\right)^2 \Omega_{s_1} + 2w_1^* w_2^* \Upsilon_{s_1,s_2}(x) + \left(w_1^*\right)^2 \Omega_{s_2} \\
&\lesssim \left(w_1^* + w_2^*\right)^2 \left(\mu^2(x) + \sigma_\varepsilon\right) + o(1) = \mu^2(x) + \overline{\sigma}_\varepsilon^2 + o(1).
\end{aligned} \tag{B.86}$$

□

Lemma B.14 (Lemma 10 - [6])**.**
For the kernel of the TDNN estimator with subsampling scales s_1 and s_2 satisfying Assumption 5 it holds that

$$\zeta_{s_1,s_2}^1(x) \sim s_2^{-1}. \tag{B.87}$$

B.6. Proof for Jackknife consistency for the TDNN estimator

Lemma B.15 (Hájek Dominance for TDNN Estimator)**.**
If $0 < \mathfrak{c} \leq s_1/s_2 \leq 1 - \mathfrak{c} < 1$ *and* $s_2 = o(n)$*, then, under Assumptions 3 and efasm:technical, the TDNN estimator satisfies the asymptotic Hájek dominance condition (Assumption 1).*

Proof. Recall the results from Lemmas B.13 and B.14.

$$\zeta_{\mathfrak{S}}^{s_2}(x) \lesssim \mu^2(x) + \overline{\sigma}_{\varepsilon}^2 + o(1) \quad \text{and} \quad \zeta_{\mathfrak{S}}^{1}(x) \sim s_2^{-1}$$

Using these results, we can find the following.

$$\frac{s_2}{n}\left(\frac{\zeta_{\mathfrak{S}}^{s_2}(x)}{s_2\zeta_{\mathfrak{S}}^{1}(x)} - 1\right) \sim \frac{s_2}{n}\left(\mu^2(x) + \overline{\sigma}_{\varepsilon}^2 + o(1) - 1\right) \sim \frac{s_2}{n} \longrightarrow 0 \tag{B.88}$$

□

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